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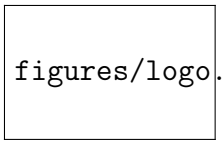

Master's Degree Course in Computer Science

Master's Thesis

Lazy Declarative Heuristics in Answer Set Programming: Design and Evaluation of a clingo Propagator

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Contents

Abstract

In the area of Answer Set Programming (ASP), ground-and-solve systems such as clingo require the instantiation of first-order rules before search begins. Although declarative domain-specific heuristics provide valuable search guidance, dynamic heuristics that depend on runtime search state (such as dynamic aggregates) suffer from combinatorial explosion during grounding. Conversely, lazy-grounding systems evaluate heuristics dynamically over partial assignments, but incur substantial search overhead because grounding and solving are interleaved throughout search. This thesis introduces a lazy declarative heuristic framework for clingo using its propagator interface, combining dynamic heuristic evaluation with the efficiency of ground-and-solve systems.

Two execution backends are provided: a compiled C++ backend with incremental aggregate state maintenance, and an embedded SWI-Prolog engine with relational query evaluation over a synchronized solver trail. Our methodology decouples heuristic representation (ground vs. lazy) from interpretation semantics (clingo-like vs. Alpha-like) through order-preserving weight flattening and the appropriate evaluation of default negation on unassigned atoms. We demonstrate that this semantic alignment is essential for dynamic heuristics to correctly steer the search. Benchmark results on three hard combinatorial problem families (the Balanced Sum Problem, the Partner Units Problem, and the House Reconfiguration Problem) show that the lazy propagator avoids grounding bottlenecks and drastically reduces memory consumption while preserving optimal search guidance. Compared to the lazy-grounding solver Alpha, our framework achieves the performance of ground-and-solve systems while maintaining the low memory cost of lazy-grounding solvers evaluating dynamic aggregates in heuristics.

Chapter 1

Introduction and Background

1.1 The Cost of a Dynamic Heuristic

Answer Set Programming (ASP) is a declarative approach to combinatorial search. Instead of describing an algorithm step by step, the modeler describes the properties that an admissible solution must satisfy, and a solver enumerates the assignments that satisfy them [1, 2, 3].

ASP solvers are essentially highly optimized backtracking, or more appropriately *backjumping*, engines. Modern ASP solving is dominated by two main paradigms: ground-and-solve and lazy grounding.

In ground-and-solve systems, such as clingo, first-order rules are instantiated over the problem domain before search starts, including the domain-specific heuristics that direct the solver's search. This approach can be severely memory-intensive, with grounding bottlenecks manifesting in two main scenarios: as problem instances scale, and whenever domain-specific heuristics depend on dynamic search state, for instance through an aggregate over the atoms that are true *at that point of the search*. Because grounding precedes search, the grounder must enumerate every possible value the aggregate could take, and a single heuristic ends up producing a number of ground objects that is combinatorial in the size of the instance. In contrast, lazy-grounding systems interleave grounding and solving. While this yields significantly higher memory efficiency, allowing a further reach on memory-intensive problems, it comes at the cost of a significantly slower search.

Depending on its underlying architecture, an ASP solver can be seen as an engine that compiles or interprets a very convenient and elegant declarative language [4] to search for a solution. The convenience of that language is paid for in control: the modeler states *what* an admissible solution is, but not *how* the engine should look for one. A pure backtracking approach behaves as an uninformed search, and on problems whose structure is known by construction, guiding the solver with that knowledge is extremely helpful.

Domain-specific heuristics are the means by which the modeler feeds that knowledge into the search. Syntactically they are declarative annotations that the grounder instantiates like ordinary rules, and their purpose is to tell the solver which atom to decide next, and with which polarity; their impact on solving is well documented [5, 6]. They are, however, part of the program, and clingo instantiates

the program before search starts. A heuristic is therefore expanded, in full, before the first decision is taken. That expansion can be relatively harmless as long as the heuristic depends only on the input. It stops being harmless as soon as the heuristic depends on *how the search is going*, which is precisely when a heuristic is most useful. Consider the directive used throughout this thesis to guide the Balanced Sum Problem (BSP), the problem of splitting $\{1, \ldots, n\}$ into two sets `b/1` and `c/1` of nearly equal sum:

```
#heuristic b(X) : x(X), not c(X), S = #sum{Y : c(Y)}, W = ((n+1)*S)+X. [W,
↪  true]
```

The rule reads: prefer to place the element X into b, the more strongly the larger the sum S of what is already in c. The aggregate value S appears *inside the weight*, and the atoms `c(Y)` it sums over are guessed during search rather than given as input facts. Since the grounder cannot know what S will be at decision time, it must enumerate every potential value of the sum ($S \in \{0, \ldots, \frac{n(n+1)}{2}\}$) and instantiate a separate ground directive for every pair (X, S). This yields $\Theta(n^3)$ ground objects (over one million (1,010,200) directives at $n = 100$) for the symmetric rules of $b/1$ and $c/1$. All of them instantiate the same single rule the modeler wrote, and all but the one matching the current partial sum are inert at any given decision. A domain-specific heuristic introduced to help the engine solve a problem quicker ends up putting the answer out of reach.

figures/diagrams/png/pipelines.png

Figure 1.1: The same heuristic under the two approaches compared in this thesis.

The alternative is to leave the heuristic symbolic and to evaluate it while the search runs, when the partial assignment that S refers to actually exists. This is the natural approach in lazy-grounding solvers such as Alpha [7, 8, 9].

1.2 Answer Set Programming

A *normal rule* in ASP has the form

$$a \leftarrow b_1, \ldots, b_m,\ not\ c_1, \ldots, not\ c_n,$$

where a, the b_i and the c_j are atoms: the head a is derived whenever every positive body atom b_i is derivable and no negative body atom c_j holds. The negation *not* is *default negation*: *not* c holds in the absence of a derivation for c. Because a rule read this way can justify itself, the meaning of a program is not given by the rules alone but through the notion of a stable model. Fix a set of atoms X, the candidate solution, and a ground program P. The *reduct* P^X is obtained by deleting every rule whose negative body intersects X, and by deleting the negative literals from the rules that survive. The reduct is a positive program, so it has a unique least model, and X is a *stable model* (or *answer set*) of P exactly when it coincides with that least model [1]. On the two-rule program

$$\begin{Bmatrix} a \leftarrow not\ b \\ b \leftarrow not\ a \end{Bmatrix}$$

the candidate $X = \{a\}$ gives the reduct $\{a \leftarrow\}$, whose least model is $\{a\}$: stable. The candidate $\{a, b\}$ deletes both rules, leaving the least model $\emptyset$: not stable. A stable model is then a set of atoms both closed under the rules and justified by them, in which every atom has a non-circular derivation that survives the model's own assumptions about what is false.

In the *guess-and-check* idiom a choice construct opens the solution space, integrity constraints prune the candidates that violate the specification, and the answer sets of the program correspond one-to-one to the solutions of the problem. ASP syntax extends normal rules with first-order variables, arithmetic, conditional literals, and three constructs this thesis uses throughout. A *choice rule* such as `{ b(X) } :- x(X).` leaves the truth values of its head atoms unconstrained, making these *choice atoms* the explicit decision points of the solver. An *integrity constraint* `:- Body.` is a rule with empty head and forbids every model that satisfies *Body*. An *aggregate atom* such as `#sum`, `#count`, `#min` or `#max` condenses a set of weighted contributions into a single comparable value [3]. Aggregates are central to this work: the heuristics under study rank decisions by expressions such as "the current sum of the elements already placed in a set", and how and *when* such an aggregate is evaluated is exactly what differentiates the systems compared here.

1.3 Ground-and-Solve: gringo, clasp, and Conflict-Driven Search

clingo [3] couples the grounder gringo with the solver clasp into a single system. The grounder instantiates the first-order program over its Herbrand universe, producing a variable-free program. To mitigate combinatorial explosion, gringo restricts this expansion to rule instances whose positive body is potentially derivable; as a result,

the ground program is generally much smaller than a full instantiation, though it must still be materialized in full before search starts. The solver clasp [10] then searches for the stable models of the ground program using *conflict-driven nogood learning* (CDNL), an algorithm that learns from conflict-inducing choices. The solver maintains a *partial assignment*, a consistent set of literals over the program atoms; it alternates *unit propagation* steps with *decision* steps, and when propagation derives a contradiction, it analyzes the conflict, learns a nogood built around the *First Unique Implication Point* (1UIP), and *backjumps* to the decision level at which that nogood becomes unit. Which atom to decide on, and with which polarity, is delegated to a *decision heuristic*; clasp's default is activity-based, in the family introduced by [11], and favors atoms that appear in recent conflicts.

This approach entails that every atom and every rule clasp will ever see must exist before the first decision is made and also that the size of the ground program is the dominant memory cost of the pipeline. The *grounding bottleneck* is a well-known limitation of ground-and-solve systems [8]; Chapter 4 measures one concrete manifestation of it, namely heuristic directives whose weight depends on an aggregate value.

1.4 Domain-Specific Heuristics in clingo

clasp allows the modeler to steer its decision heuristic using domain-specific knowledge through the `#heuristic` directive [5]:

```
1 #heuristic A : Body. [W@P, M]
```

Here, A is the target domain atom whose decision choice is steered according to the modifier M (e.g., setting its sign or priority level). The heuristic applies only if the rule *Body* is satisfied. If multiple heuristic directives target the same atom A, clingo selects the one with the highest priority P to determine the atom's modifier and weight. Across distinct target atoms, clasp then prioritizes candidates with higher weight W. In contrast, Alpha treats (P, W) as a global lexicographic rank over all atoms (Section 1.5).

1.5 Lazy Grounding, Alpha, and Heuristics over Partial Assignments

Alpha is the reference state-of-the-art lazy-grounding solver for this work.

Comploi-Taupe et al. extended Alpha with declaratively specified domain-specific heuristics evaluated against the *partial* assignment maintained during solving [8]. Its semantics differs from clingo's on two main points.

Specifically, Alpha evaluates negated body literals against unassigned atoms and prioritizes P over weight W in the tuple $(W@P)$, always preferring higher priority values. Only when two atoms, even different, have the same priority are they compared by weight. A complete formalization of these semantic differences and their integration into our experimental design is provided in Chapter 2.

1.6 Extending clingo: Propagators and the Heuristic Interface

Standard clingo does not evaluate heuristic conditions dynamically at search time. However, it provides an extension mechanism: *theory propagators*, user-defined components registered with the solver that participate directly in the search loop [12]. This thesis leverages that interface to provide search guidance computed dynamically on the trail rather than fully materialized before search.

A propagator has four callbacks. During `init`, it inspects the symbolic atoms of the ground program, maps the atoms it cares about to solver literals, and registers *watches* on them. During search, `propagate` is invoked when a watched literal becomes true, `undo` when backtracking retracts it, and `check` on total assignments. The propagator is invoked by clasp only when a watched literal changes state.

The `clingo::Heuristic` interface inherits from Propagator and extends it with a `decide` callback, invoked at each decision step *before* the solver's built-in heuristic. The Heuristic object receives the current assignment of the solver and either returns a signed literal, which becomes the next decision, or returns 0, in which case clasp falls back to its default activity-based heuristic (VSIDS [11]). This is the mechanism the rest of the thesis builds on. It keeps the declarative heuristic descriptions in memory, evaluates them incrementally against the trail, and answers each `decide` call with the best applicable target.

1.7 Objective and Contributions

This work asks whether the same effect of lazily evaluating heuristics can be obtained inside clingo without giving up its ground-and-solve pipeline and speed advantage, that is, whether a ground-and-solve engine can be equipped with lazily evaluated heuristics (Figure 1.1) and whether the overhead of dynamic evaluation still allows the engine to outperform a state-of-the-art lazy-grounding solver.

The thesis makes the following contributions.

1. **A lazy heuristic evaluator for clingo:** A propagator, registered through the public `clingo::Heuristic` interface, that keeps a declarative heuristic in memory as a compact description and evaluates it at every decision against the current partial assignment.
2. **Two propagator backends:** A compiled C++ backend, which materializes one candidate per ground target and maintains incremental aggregate states, and an in-process SWI-Prolog backend, which evaluates the heuristic as an ordinary logic program against a mirrored image of the solver trail.
3. **Semantics and ablation study:** Ground-and-Solve, Lazy, clingo semantics and Alpha semantics combined give four variants of every encoding plus a heuristic-free baseline, so that no measured difference has to be attributed to two causes at once.
4. **Weight translation from Alpha's syntax:** Alpha's priority levels are global, while clingo's are local to a single atom. Section 2.1.3 presents the transformation

that folds a level into the weight, the conditions under which it preserves the intended order and the weight field limit imposed by clingo.

5. **Trade-off analysis of the lazy approach.** By collecting per-decision metrics in both backends, we evaluate the exact computational overhead of lazy grounding against its savings in program size.

The evaluation covers three problem families, the Balanced Sum Problem, the Partner Units Problem, and the House Reconfiguration Problem, on both backends, with a correctness harness that checks that no variant changes the answer sets. The implementation is available at[1].

[1] `https://github.com/pierspad/clingo-lazy-heuristic-evaluator`

Chapter 2

Methodology

2.1 The Experimental Design: Two Axes, Four Variants

Every experiment in this thesis compares encodings of the same problem that differ in how the domain-specific heuristic is represented and interpreted. The rules that define the problem, its guess and its constraints, are held fixed across the variants of a family; what varies are the two properties that Section 1.4 and Section 1.5 identified as the points on which clingo and Alpha disagree, and they vary independently.

The *approach* axis fixes where the heuristic is evaluated. In the ground-and-solve approach the heuristic is written with native `#heuristic` directives: gringo expands them before search and clasp's Domain heuristic consumes the resulting ground objects. In the lazy approach the heuristic remains a handful of ground facts, `__heuristic(...)` for the native backend and `heuristic("...")` for the Prolog one, which the propagator reads at its initialization and evaluates at every decision step against the partial assignment.

The *semantics* axis fixes how a heuristic condition reads partial information: the clingo-like semantics, under which `not a` holds only once a has been assigned false and an aggregate has a value only once its sources are determined, against the Alpha-like semantics, under which `not a` holds as soon as a is not assigned true, and an aggregate is computed over the atoms currently true. Only the second reading makes a heuristic dynamic in the sense of reacting to a partial assignment during search, as motivated in Section 1.1: an instruction such as "prefer the element whose insertion keeps the two sums balanced" presupposes that the sum can be evaluated against an assignment that is still being built; under clingo's semantics, that sum simply has no value until it is too late for the guidance to matter.

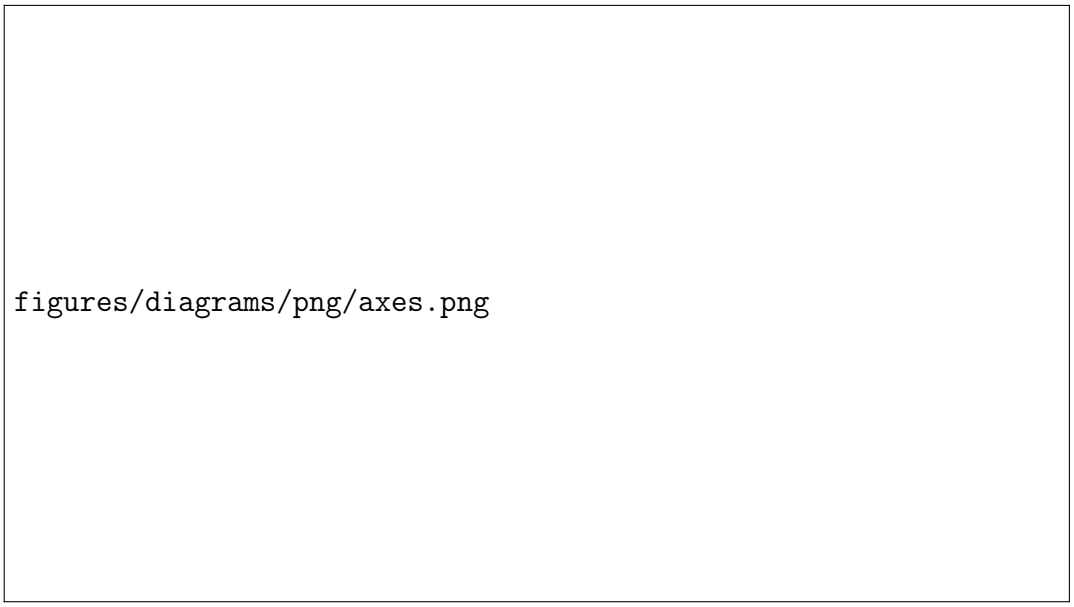

Figure 2.1: The experimental matrix. Prefixes name the approach, suffixes the semantics.

Crossing the two axes yields the four variants that the rest of the thesis refers to by: `gc` and `ga` for the ground approach under the clingo and the Alpha semantics, `lc` and `la` for the lazy evaluated one, the prefix naming the approach and the suffix the semantics. To them is added the heuristic-free baseline `gc_noheur`, which carries the same domain logic with no heuristic at all and measures what the solver does when it is left alone (Figure 2.1).

To isolate specific architectural and modeling mechanisms beyond this primary matrix, the experimental design also incorporates three targeted exploratory variants: `ga_weak`, `la_aux`, and `la_co`. Their empirical motivation and methodological role in disentangling the contributions of default negation, body materialization, and constraint formulation are discussed in Section 2.1.2.

2.1.1 Simulating Alpha semantics

Reproducing an Alpha-like behaviour within clingo requires addressing two distinct aspects: default negation over unassigned literals and dynamic aggregate evaluation.

Default negation over unassigned literals

The first transformation concerns negative literals in heuristic bodies: `not c(X)` in BSP, `not not_assigned_sensor_unit(S,U)` in PUP, and guards such as `not fullCabinet(C)` and `not assignedThing(T)` in HRP.

Clasp applies heuristic modifications only to variables that are currently unassigned, as enforced in `DomainHeuristic::propagate`:

```
1 if (s.value(a.var) == value_free && a.prio >= prio) { applyAction(s, a, prio);
  ↪ }
```

It follows that *a directive whose body entails the truth value of its own target atom can never fire*: once the body conditions hold, the target is already decided, and clasp skips it.

All heuristic directives in the suite that test the search state through negative literals show the same problem:

- In **BSP**, `b(X)` and `c(X)` are complementary choices, so the condition `not b(X)` is satisfied only once the target `c(X)` is already derived.
- In **PUP**, the condition on `assigned_sensor_unit(S,U)` in the heuristic is identical to the problem rule that derives that same atom.
- In **HRP**, the placement directives for `cabinetTOthing(C,T)` require `not assignedThing(T)`; in ground clingo this holds only when the candidate assignments for T are already proven false, meaning the target is never unassigned (`value_free`) when the directive evaluates.

Consequently, in the `gc` variant, none of these ever fires during search.

The ground variant `ga` therefore drops these negative literals from the heuristic bodies. For directives where the negated literal simply tests the target atom or its complement (as in BSP and PUP), this simplification is completely safe: nothing is lost that could have fired, and clasp's `value_free` check still prevents modifying an already assigned atom. In HRP, however, the dropped literals also include capacity guards such as `not fullCabinet(C)`, which describe a shared resource rather than the target atom itself. There, dropping the guard is a necessary static approximation: it enables the directive to fire on unassigned atoms from the start of search, at the price of never deactivating once a cabinet becomes full.

Dynamic aggregates by bound-guarded unrolling

The second transformation replaces dynamic aggregate equality with bound-guarded unrolling over all candidate values. For BSP:

```
1 sum_value(0..tot).
2 #heuristic b(X) : x(X), sum_value(S), S <= #sum{Y : c(Y)},
3     W = ((n+1)*S)+X. [W, true]
```

Here the directive is unrolled over all `sum_value(S)` and guarded by a bound condition, so that during search the currently satisfied bounds encode the current aggregate value, provided the guard is monotone, i.e. once satisfied it stays satisfied as further atoms become true.

Without explicit priorities all unrolled directives share the default priority 0. Clasp applies newly satisfied directives by overwriting previous ones; it therefore retains the *most recently activated* directive, not the maximum-weight one. To match the reference heuristic, the unrolling must align the activation order with the weight order:

- **Lower-bound guards ($V \leq f$):** As f increases, the active set $\{V : V \leq f_{curr}\}$ expands upwards. The last activated value is $V = f_{curr}$, which yields the maximum weight if $W(V)$ is strictly *increasing* (e.g. in BSP, where $W(S) = (n{+}1)S + X$).

- **Upper-bound guards ($V \geq f$):** The active set $\{V : V \geq f_{curr}\}$ instead expands downwards, as f decreases, and the alignment requires $W(V)$ to be strictly *decreasing*.

2.1.2 Methodological Rationale for the Exploratory Variants

While the four main variants evaluate the core matrix of approach versus semantics, three targeted exploratory variants are introduced as experimental controls to isolate specific factors:

Isolating Alpha-Negation from Dynamic Aggregates (`ga_weak`). Alpha-like semantics treat default negation as satisfied over unassigned literals, and evaluate aggregates dynamically over currently true atoms. While `ga` simulates both via extensive ground unrolling, `ga_weak` acts as an ablation study that applies Alpha's negation semantics while avoiding unrolling aggregates. This investigates whether adapting default negation alone provides any improvement during search, or whether dynamic aggregate evaluation is strictly indispensable for an effective search guidance.

Deconstructing Lazy Efficiency: Directives vs. Body Materialization (`la_aux`). Lazy evaluation replaces thousands of ground `#heuristic` directives with a minimal set of declarative facts. This raises a natural question: does the performance gain come merely from having fewer heuristic directives, or from keeping the calculation of heuristic conditions entirely out of the ground program? The `la_aux` variant tests this distinction by pre-materializing all heuristic conditions into auxiliary ASP rules (`h_b(X,S)`), while querying them through the lazy propagator.

Disentangling Heuristic and Problem Constraint Bottlenecks (`la_co`). In the baseline BSP encoding without heuristics, the balance constraint generates a combinatorial explosion of ground rules. Consequently, even though lazy evaluation reduces search time, total execution remains dominated by grounding the base problem constraints. The `la_co` variant combines the lazy evaluation with a compact linear formulation of the balance constraint.

2.1.3 Weights, Priorities, and Flattening

Section 1.5 recorded that the two systems read the annotation $[W@P]$ of a heuristic differently. In Alpha, the pair is an *absolute* rank over all heuristic instances, read lexicographically as (P, W): the priority P is a global level and decides first, so no instance of level P is considered while an instance of a level $P' > P$ is still applicable, and the weight W only orders the instances that share the same level. In clingo, the evaluation is different: P is local, arbitrating only between heuristics relative to the *same* target atom, while the order in which distinct atoms are decided comes from the weight alone. An encoding written for Alpha therefore cannot be handed to clingo verbatim.

The intended order can nevertheless be recovered, by *flattening* the level into the weight:

$$W_{flat} \;=\; W + P \cdot M, \qquad M > \max W - \min W,$$

where the intra-level weights W range over $[\min W, \max W]$ and the level multiplier M is larger than that range. Under this condition $W + P \cdot M > W' + P' \cdot M$ holds whenever $P > P'$, regardless of the two individual weights, and reduces to $W > W'$ when $P = P'$: the total order induced on the atoms is then the one Alpha's absolute levels induce. When all weights are non-negative, which is the case in every encoding of this suite, $M > \max W$ suffices.

Each benchmark fixes its own M. In the PUP encodings M is the constant offset 100,000 added to the zone weights, against a maximum sensor weight of 26,220 on the benchmark instances; in HRP M is derived from the instance as `levelMul(LM) :- LM = MC + MR + 10` over the maximum cabinet and room identifiers, which gives 16 on the smallest instance and 54 on the largest. BSP does not need a multiplier, as discussed in Section 3.3.1.

2.1.4 Problems on Bigger Weights

clasp stores the weight of a `#heuristic` modification in a signed 16-bit integer field and clamps any value exceeding 32,767. Consequently, two ground directives whose weights both exceed 32,767 become indistinguishable to clasp's Domain heuristic. A flattened weight must therefore satisfy $W + P \cdot M \leq 32{,}767$ for the ground variants to rank as intended. HRP stays far below this bound; PUP exceeds it on the zone directives, with the consequences discussed in Section 4.10; and BSP, whose maximum weight grows as $\Theta(n^3)$, crosses it within the benchmarked range. What matters there is not the largest weight the grounder emits, but the largest one the search actually reaches: directives whose aggregate value is never attained may clamp without any observable effect. On BSP that effective crossing occurs at $n = 51$, as analyzed in detail in Section 4.7. The lazy backends are not subject to this bound because they rank targets internally and hand clasp a ready decision rather than a weight.

The suite adopts clingo's convention in *all four variants and in both backends*: absolute levels always live in the weight, simplifying the propagator as it does not need to account for a separate priority field. Once global levels are folded into weights, weight becomes the sole ordering criterion under both semantics. Accordingly, the native lazy language (Section 2.3) omits the priority field entirely.

A last convention concerns empty aggregates: `#max` over an empty set is 0 in Alpha but $-\infty$ (`#inf`) in clingo. All encodings therefore write `#max{0; ...}` with an explicit neutral element, and both lazy backends implement the same rule (the native `MaxState` returns 0 when empty; the Prolog runtime defines `dyn_max` with a 0 default).

2.2 System Architecture

The project maintains two modified copies of clingo. They share the same upstream sources and differ only in their extensions: the build in `clingo-native` carries the

C++ backend, while the build in `clingo-prolog` carries the Prolog backend, configured with `clingo_USE_SWIPL=ON` so that the SWI-Prolog engine is linked in-process, making its memory impact easier to account for. In both builds, the extension lives entirely under `libclingo`, exposes the same class name, and is registered at the same point of `clingo_app.cc`:

```
HeuristicPropagator my_propagator;
ctl.register_propagator(my_propagator, true);
```

The propagator is registered on *every* run of the modified binary. On a program that contains no lazy heuristic facts, the propagator itself is inactive: `init` finds nothing to watch and `decide` always returns 0, so every decision is left to clasp's own branching heuristic. The same binary therefore also runs the ground variants (with clingo's standard `#heuristic`s) and the heuristic-free baseline without requiring a separate build.

The second argument, `true`, instructs clasp to call the propagator sequentially, avoiding race conditions on the incremental state described in Section 3.1.1. No other parts of clingo are modified. Figure 2.2 shows the architecture of the two binary build variants.

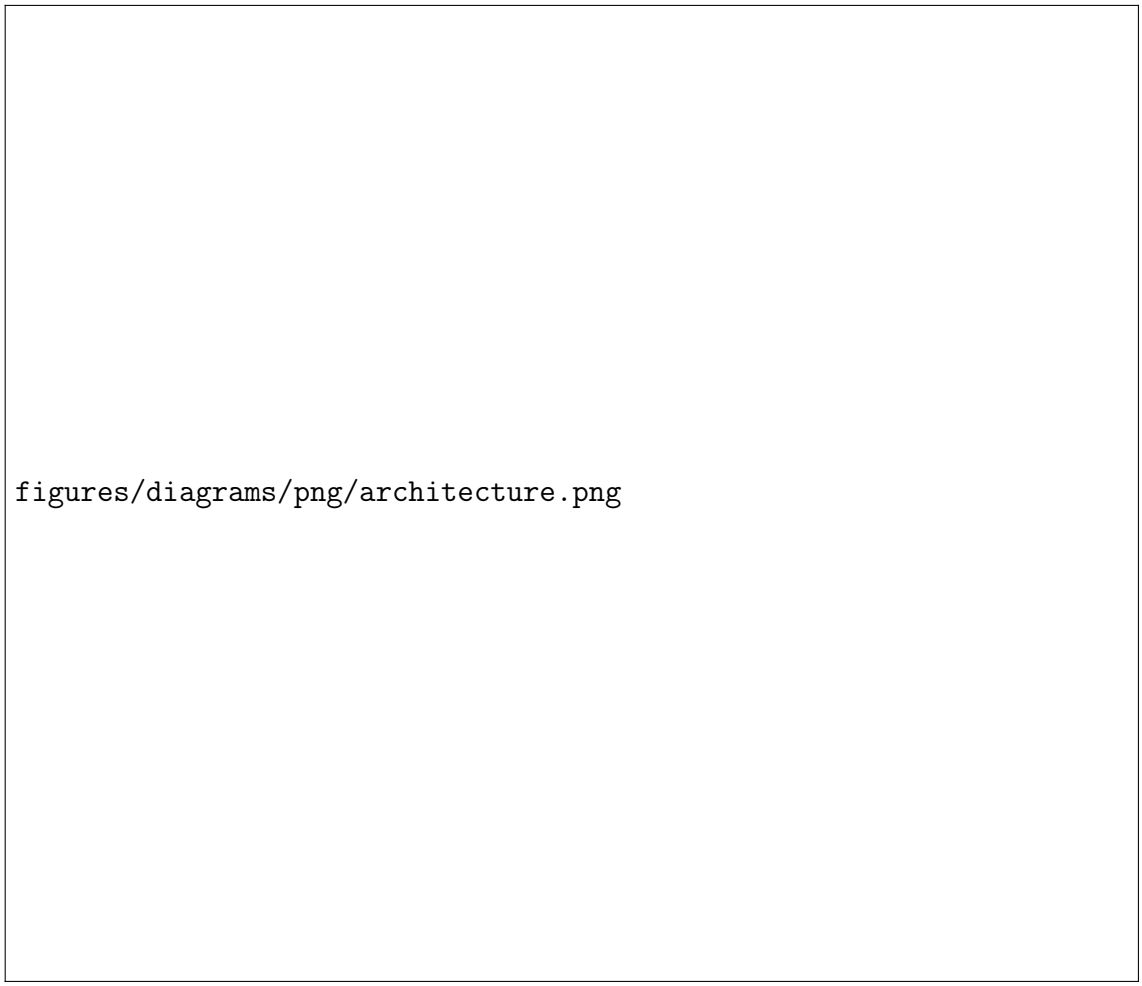

Figure 2.2: System architecture of the two binary variants, showing their respective heuristic propagators and evaluation backends attached to clasp.

In both builds, the class implements `clingo::Heuristic`,the propagator interface extended with the `decide` callback (Section 1.6). Both backends follow the same four-step lifecycle:

- During `init`, they scan symbolic atoms for trigger facts (`__heuristic(...)` in the C++ build and `heuristic("...")` in the Prolog build), construct their runtime representation, and register watches on solver literals.
- During forward search, `propagate` updates the internal state of the aggregates appearing in the heuristic bodies, together with the candidate states, as watched literals become true.
- During backtracking, `undo` restores those aggregate and candidate states as literals are retracted from the trail.
- At each decision point, `decide` either returns a signed literal for the highest-ranked target or returns 0, delegating the decision to clasp's fallback heuristic.

The two backends are not symmetric in what they can express. The native backend maintains an incremental state for `#sum`, `#count`, `#min` and `#max` over one predicate

and one argument position, and can only filter the aggregated set by matching its arguments with those of the target atom; the Prolog backend evaluates an arbitrary `aggregate_all/3` goal and is unrestricted in this respect. Section 2.3 states the fragment each backend accepts, and what falls outside it.

To activate this mechanism, runs are launched with the flag `-heuristic=Domain`. A detailed discussion of the architectural differences between the two backends is provided in Chapter 3.

The propagator never adds a constraint, never derives a literal, and never refuses one. It only reorders the decisions clasp would have taken anyway, so the answer sets are untouched, and a heuristic can make a run slower or faster but cannot change the meaning of the encoding (detailed in Section 3.1.3).

2.3 Translation of the Heuristic Directives

Neither backend reads native `#heuristic` directives. Instead, a custom declarative "lazy heuristic" language replaces them with a single ground fact that *describes* the rule. This section presents what one may write, and what each construct means, by deriving that fact for the BSP heuristic of Section 1.1 one component at a time.

The starting point is the directive as clingo reads it:

```
#heuristic b(X) : x(X), not c(X), S = #sum{Y : c(Y)}, W = ((n+1)*S)+X. [W,
↪  true]
```

Target. The target atom is `b(X)`, where X ranges over the ground instances. The lazy fact specifies only the predicate name, `__target(b)`. During `init`, the propagator queries clingo's atom table for all ground instances of `b(...)` already materialized by gringo.

Positive body. The condition `x(X)` shares its argument with the target: the candidate for `b(7)` is alive only while `x(7)` is true. A condition of this shape is written as the bare predicate name, `x`, and the propagator matches its arguments against the target tuple position by position.

However, a body predicate may have a different arity; its arguments may sit in different positions; or a value may have to be read out of it and used in the weight. The correspondence is then stated explicitly:

```
__body(pred, __match(src, tgt), __bind_arg(var, idx))
```

`__match(src, tgt)` requires the argument at the `src` position of the body atom to equal the argument at the `tgt` position of the target atom, where both `src` and `tgt` are integers. All indices are 0-based, and at least one `__match` is mandatory. `__bind_arg(var, idx)` extracts argument `idx` of the body atom into a variable.

Given the ground facts:

```
score(1,10). score(2,5). score(3,7).
```

and the target atoms:

```
choose(1). choose(2). choose(3).
```

the directive:

```
__heuristic(__target(choose),
            __body(score, __match(0,0), __bind_arg(w,1)),
            __weight(w), __modifier(true)).
```

pairs each `choose(X)` with the `score(X,W)` that agrees with it on the first argument, and binds `w` to the second argument of that atom. The weights are then 10, 5 and 7, so the propagator decides `choose(1)` first.

Negative body. The condition `not c(X)` likewise matches its arguments against the target tuple. Negative body literals are specified by prefixing the predicate name with `__n_`.

Semantics. The reading of that negation is fixed by the `__semantics(S)` selector, which decides whether `__n_c` means "`c(X)` is assigned false" or "`c(X)` is not assigned true". The same selector fixes the reading of aggregates, described below. It takes `alpha` or `clingo`, and defaults to `clingo`, so that a directive written without it behaves as the native `#heuristic` it replaces.

Aggregates. Aggregates are evaluated lazily during search. `__bind(s, __sum(c))` binds variable `s` to the sum over the predicate `c` (by default aggregating its first argument). The operators `__count`, `__min`, and `__max` follow the same structure.

An aggregate can be restricted *per target* using `__filter(src_idx, tgt_idx, offset)`, which keeps only source atoms whose argument at position `src_idx` equals argument `tgt_idx` of the target plus an optional `offset`. For instance, in PUP, retrieving the maximum number of sensors on the *previous* unit $(U-1)$ for a target `assigned_zone_unit(Z,U)` is expressed as:

```
__bind(m1, __max(num_sensors_on_unit, 0, __filter(1, 1, -1)))
```

where `0` selects the aggregated argument N in `num_sensors_on_unit(N, U')`, and the filter matches $U' = U - 1$.

Weight. Weight is expressed as a *term tree* such as:

```
__weight(__add(__mul(s, B), self))
```

which represents $(s \cdot B + \mathit{self})$, where `self` is a shorthand for the target atom's argument at index 0 without requiring a variable binding.

The binary constructors `__add`, `__sub`, and `__mul` form expression trees whose leaves belong to three categories: integer constants, variables of the heuristic language, and target arguments (accessed through `self` or `__bind_target_arg(var, idx)` for any other).

At `init`, the propagator compiles all weight term trees into ASTs. During search, each active heuristic object evaluates its weight AST dynamically against the current trail state, avoiding the memory overhead of materializing them in the ground program. Only candidates with a strictly positive resolved weight are taken into account; the others are ignored, leaving their decision to the solver's default heuristic.

Modifier. The `__modifier(M)` selector specifies which of clingo's heuristic modifiers to use: `level` sets the ranking value alone, `sign` sets the preferred polarity alone, while `true` (the default) and `false` set both.

The solver's `decide` interface expects a *signed* literal, so a ranking value alone does not determine a decision. Each target keeps one slot per channel, ranking value and preferred polarity, and when several directives write to the same slot the one with the highest weight wins. This is what replaces clingo's local priority. Across the benchmarks all directives use `true`, except for the final layer of HRP, which uses `false` to prune the remaining choice atoms.

Assembling the components gives the fact in full:

```
__heuristic(__target(b), x, __n_c, __bind(s, __sum(c)),
            __weight(__add(__mul(s, B), self)),
            __semantics(alpha), __modifier(true)) :- B = n + 1.
```

construct	clingo directive	native lazy fact	Prolog rule string
structure			
target	`b(X)`	`__target(b)`	`heuristic(b(X), W, P, M)`
positive body, target tuple	`x(X)`	`x`	`holds(x(X))`
positive body, join	`p(X,V)`	`__body(p, __match(0,0), __bind_arg(v,1))`	`holds(p(X,V))`
negation	`not c(X)`	`__n_c`	`alpha_not(c(X))` `clingo_not(c(X))`
aggregate	`S = #sum{Y : c(Y)}`	`__bind(s, __sum(c))` also `__count`, `__min`, `__max`	`aggregate_all( sum(Y), holds(c(Y)), S)`
filtered aggregate	`M = #max{N : q(N,U-1)}`	`__bind(m, __max(q, 0, __filter(1,1,-1)))`	`U1 is U-1, dyn_max( holds(q(N,U1)), N, M)`
target arguments	`X`, `Y` in `b(X,Y)`	`self`, `__bind_target_arg( y, 1)`	ordinary head variables
annotations			
weight	`W = ((n+1)*S)+X`	`__weight(__add( __mul(s,B), self))`	`W is B*S+X`
modifier	`[W, true]`	`__modifier(true)`, default `true`	fourth head argument
semantics	, (fixed by the system)	`__semantics(alpha)` `__semantics(clingo)`	choice of `alpha_not` / `clingo_not`
target still free	, (implicit in clasp)	implicit: an assigned target is skipped	`target_available( b(X))`

Table 2.1: The same heuristic in the three notations.

Chapter 3

Implementation and Benchmark Setup

3.1 Inside the Native Backend

3.1.1 Parser and Runtime Representation

The parser in `libclingo/src/heuristic_parser.cc` turns each `__heuristic` symbol into a `HeuristicRuleTemplate`: target atom, positive and negative body literals, variable bindings, modifier, semantics, and an arithmetic AST for the weight.

During `init`, the propagator inspects the ground symbolic atoms and builds an index mapping each predicate and its argument tuple to the corresponding solver literal. It then materializes candidates: for each ground target atom and matching combination of positive body atoms, it creates a `RuntimeHeuristicCandidate` storing the target literal, target tuple, body literals, bound variable values, and instantiated aggregate keys.

In BSP with $n = 100$, this yields only 100 candidates for `b(X)` (and 200 across both partitions), whereas the standard ground encoding generates 1,010,200 directives ($2 \times 100 \times 5051$) to cover all possible discrete sums $S \in \{0, \ldots, 5050\}$. Thus, candidate materialization in the lazy propagator scales linearly with the number of ground targets rather than with the domain size of dynamic aggregates, eliminating exhaustive grounding.

Finally, the propagator registers watches exclusively for literals that affect heuristic state: both polarities of target and negative body literals, positive body literals, and aggregate source literals.

3.1.2 Incremental Aggregates

Aggregate states implement three methods: `add`, `remove`, and `result`.

Sums and counts (`SumState`, `CountState`) store a single `int`, because adding and removing values can be done with simple addition and subtraction. In contrast, min and max (`MinState`, `MaxState`) cannot be undone with arithmetic; when the current minimum or maximum is removed during backtracking, the state must know what the next best value was. They therefore use an `std::map<int, int>` that maps

each active value to how many times it occurs.

Each source atom updates the `AggregateState` matched by its `RuntimeAggregateKey`: values are added in `propagate` and removed in `undo`.

Under *clingo* semantics, an aggregate is evaluated only after all its source literals are assigned, gating dependent candidates until then. Under *Alpha* semantics, candidates directly read whatever partial aggregate value is currently available on the trail.

3.1.3 Evaluation and Decision

A candidate is *active* when its target literal remains unassigned, all its positive body literals evaluate to true, and every negative body literal is satisfied under the template's semantics. The two supported semantics differ only in how they treat negative literals:

```
static bool negative_literal_is_satisfied(HeuristicSemantics s,
↪  clingo::TruthValue v) {
    return s == HeuristicSemantics::clingo ? v == clingo::TruthValue::False
                                           : v != clingo::TruthValue::True;
}
```

Once a candidate satisfies its body conditions, the propagator computes its numerical weight by evaluating its weight AST over the current aggregate values and term arguments. The resulting effect is then applied to the target literal's `TargetHeuristicState`.

The native propagator adopts four of clingo's domain heuristic modifiers:

- `level`: sets the branching priority to w (higher values are chosen first), leaving the truth polarity unchanged;
- `sign`: sets the preferred truth polarity based on the arithmetic sign of w (positive for *true*, negative for *false*), using $|w|$ to resolve precedence among conflicting sign directives;
- `true`: sets the branching priority to w and forces the truth polarity to *true*;
- `false`: sets the branching priority to w and forces the truth polarity to *false*.

clingo's remaining modifiers, `init` and `factor`, are not supported because custom propagators cannot manipulate clasp's internal VSIDS score tables.

If multiple heuristic rules target the same atom, their effects are combined into a `TargetHeuristicState` that tracks the `level` and `sign` across all active `RuntimeHeuristicCandidate` instances.

Unassigned targets with an active `level` are stored in an ordered set:

```
std::set<DecisionRankKey, DecisionRankGreater> active_decision_ranks_;
```

Entries are sorted by decreasing weight. Because weights are stored as standard 32-bit integers, this avoids the 16-bit saturation limit of clasp's internal domain table (Section 2.1.3).

When clasp calls `decide`, the propagator inspects `active_decision_ranks_` from the beginning, discarding any targets that are already assigned on the trail or whose weights have changed. It then branches on the first valid target using its stored `sign`. If the set is empty, the propagator checks whether clasp's fallback atom has a stored `sign`; otherwise, it returns 0 to delegate the decision to clasp. During backtracking, `undo` restores aggregates and candidate states using `noexcept` functions to prevent state corruption during solver unwinding.

3.2 The Prolog Backend

While the native backend prioritizes evaluation speed through compiled C++ structures, the Prolog backend evaluates heuristics as arbitrary logic programs via an embedded engine, trading runtime overhead for unrestricted expressiveness. This section describes its runtime architecture and trail synchronization mechanisms (with a comprehensive performance comparison presented in Section 4.13).

3.2.1 Rule Strings and Normalization

In the Prolog backend, a heuristic is defined as a rule string whose head is a four-argument term: `heuristic(Target, Weight, Priority, Modifier)`. The BSP heuristic from Section 2.3 is specified as follows:

```
heuristic("heuristic(b(X), W, 0, true) :-
    aggregate_all(sum(Y), holds(c(Y)), S), holds(heuristic_base(Base)),
    holds(x(X)), target_available(b(X)), alpha_not(c(X)),
    W is Base*S+X.").
```

The rule body uses standard Prolog syntax and maps directly to the native constructs: `holds/1` for positive conditions, `alpha_not/1` for negative conditions, `aggregate_all/3` for dynamic sums, and `is/2` for arithmetic evaluation of weights. Furthermore, `target_available/1` ensures the target atom is still unassigned. While the native backend checks this implicitly during candidate selection, the Prolog rule must state it explicitly to prevent proposing atoms that are already assigned on the trail.

During initialization, the propagator collects all `heuristic/1` facts, normalizes the rule strings, and classifies their semantics: rules containing `clingo_not(...)` use clingo semantics, while all others use Alpha semantics. The propagator extracts the predicate signatures referenced in the rules (`holds`, `alpha_not`, `clingo_not`, `target_available`, and rule targets) and asserts into Prolog only solver atoms matching those signatures, avoiding unnecessary memory and synchronization overhead from irrelevant predicates.

3.2.2 The Runtime Program

During initialization, the backend starts an embedded SWI-Prolog engine [13] and loads a set of base Prolog rules:

```
holds(A) :- true_atom(A).
holds(A) :- static_atom(A), \+ true_atom(A).
clingo_not(A) :- false_atom(A).
alpha_not(A)  :- \+ holds(A).
target_available(A) :- target_atom(A), \+ true_atom(A), \+ false_atom(A).
dyn_max(Goal, Template, Max) :-
    (aggregate_all(max(Template), Goal, R) -> Max = R ; Max = 0).
```

As clasp makes decisions, the propagator mirrors the trail in the Prolog database by adding `true_atom(A)` when an atom is assigned to true and `false_atom(A)` when it is assigned to false, removing them upon backtracking. Undecided atoms simply do not exist in the database. Because of this, `alpha_not(A)` succeeds whenever an atom does not hold true. In contrast, `clingo_not(A)` requires an explicit `false_atom(A)` on the trail. In the same way, `aggregate_all` dynamically computes sums and counts over the atoms currently true on the trail. `target_available(A)` verifies that a target is neither true nor false before branching on it, and `dyn_max/3` defaults to 0 when evaluating an empty set (Section 2.1.3).

3.2.3 Synchronization and Decision

Trail synchronization is incremental: during search, `propagate` and `undo` update the state of atoms in the Prolog database as their solver literals change on the trail.

At each decision point (`decide`), the backend queries all solutions to `heuristic(T, W, P, M)`, filters out targets that are already assigned on the trail, and branches on the candidate with the highest priority and weight. Unlike the native C++ backend, which incrementally maintains a sorted set of active targets, the Prolog backend re-evaluates the heuristic query from scratch at every decision step. This design prioritizes relational expressiveness over per-decision efficiency (evaluated in Section 4.6).

3.3 Benchmark Problems and Encodings

3.3.1 Balanced Sum Problem (BSP)

The BSP divides the set of integers $\{1, \ldots, n\}$ into two disjoint sets, `b/1` and `c/1`, such that their sums differ by at most one. It is the motivating example of [9], and the encoding used here as the Alpha reference is the authors' own `Evaluation/Motivating_Example/Balanced_Sum_Alpha.asp`, taken unmodified from the supplementary material of that paper (Section 3.3.4). The heuristic greedily assigns unassigned integers to set b with a priority proportional to the current sum of set c, driving the solver toward balanced partitions. Because all decisions share a single priority level, the current sum S acts as the primary ranking term, with the element value X resolving ties between equal sums.

The implementation of the three exploratory variants introduced in Section 2.1.2 on BSP required some specific adjustments:

- `ga_weak`: The ground heuristic directives omit the negative body guards (`not c(X)` / `not b(X)`), retaining only the static aggregate binding `S = #sum{Y : c(Y)}` without unrolling over candidate values:

```
#heuristic b(X) : x(X), S = #sum{Y : c(Y)}, W = ((n+1)*S)+X. [W, true]
#heuristic c(X) : x(X), S = #sum{Y : b(Y)}, W = ((n+1)*S)+X. [W, true]
```

  Dropping the `sum_value(S)` domain predicate does not reduce the grounding: the equality binding still forces gringo to enumerate every reachable sum, so `ga_weak` materializes the same $\Theta(n^3)$ directives as `gc` and `ga`. What changes is when they fire, since the equality holds only once every atom of the sum is assigned (Section 4.7).

- `la_aux`: Heuristic conditions and aggregate values are pre-materialized:

```
h_b(X,S) :- x(X), not c(X), S = #sum{Y : c(Y)}.
h_c(X,S) :- x(X), not b(X), S = #sum{Y : b(Y)}.
```

  The lazy propagator language is then instructed to monitor these auxiliary atoms via the `__body` constructor:

```
__heuristic(__target(b), __body(h_b, __match(0,0), __bind_arg(s,1)),
            __weight(__add(__mul(s, B), self)),
            __semantics(alpha), __modifier(true)) :- B = n + 1.
```

  The propagator will track truth-value assignments over the ground `h_b(X,S)` atoms on the trail, testing whether auxiliary trail synchronization reintroduces solver overhead.

- `la_co`: uses the same lazy heuristic as `la`, but instead of computing and comparing the two sums separately:

```
sumB(Sum) :- Sum = #sum{Y : b(Y)}.
sumC(Sum) :- Sum = #sum{Y : c(Y)}.
:- sumB(SB), sumC(SC), (SB - SC)**2 > 1.
```

  it enforces the partition balance directly with a single aggregate rule:

```
#const min_val = tot / 2.
#const max_val = (tot + 1) / 2.
:- not min_val <= #sum{ Y : b(Y) } <= max_val.
```

  This removes the heavy grounding bottleneck of the base problem while keeping the meaning of the heuristic unchanged.

3.3.2 Partner Units Problem (PUP)

The PUP [14] assigns communication units to interconnected zones and sensors under capacity and topology constraints. The heuristic uses a two-level priority policy: zones are assigned first, followed by sensors.

The base encoding is taken from the Alpha reference literature [14, 9]: the reference file is the authors' `Evaluation/PUP/Double/encoding-alpha-prologheu-double.asp`, and the `double-`N instances are byte-identical to theirs, so both systems run on the same files. However, because the original encoding was written for a lazy solver, running it directly in clingo causes a severe grounding bottleneck. To mitigate this while preserving exact solution equivalence, the encoding was refactored using standard ASP modeling practices: bounding unit indices to valid ranges, expressing capacity limits via linear `#count` aggregates, and eliminating a redundant blocking rule that generated excessive ground constraints.

Finally, the global priority of zones over sensors is flattened into the heuristic weights using the +100,000 offset introduced in Section 2.1.3.

3.3.3 House Reconfiguration Problem (HRP)

The HRP [15] places items into cabinets and assigns cabinets to rooms under person-ownership and dimensional constraints, minimizing deviations from an existing legacy setup. The heuristic enforces a five-level priority policy: reusing legacy components, placing long items, placing short items, assigning cabinets to rooms, and finally setting unassigned choice atoms to false. It is the heuristic of Case Study 1 of [8]; the reference file is the authors' `DomainHeuristics/House/house_alpha_2021-02-04.asp`, and the encodings of this suite are a rule-by-rule port of it.

Unlike BSP and PUP, the HRP heuristic uses no dynamic aggregates: its conditions are ordinary literals and its weights plain arithmetic, and the aggregates of the encoding sit one rule further away, in the definitions of `fullCabinet/1` and `fullRoom/1`. HRP therefore isolates the behavior of lazy default negation (`la` vs. `lc`) on partial assignments.

3.3.4 Alpha as an External Reference

We benchmark our propagator against the lazy-grounding solver Alpha [7, 8] as an external state-of-the-art reference. Specifically, we test the *Qh* (Query Heuristics) variant developed by Comploi-Taupe et al. [9], invoked with `-uqh`, which evaluates domain heuristics and dynamic aggregates over partial assignments via Prolog queries. This mechanism directly mirrors our Prolog backend and implements the semantics our native propagator reproduces.

The choice of build matters: the official Alpha 0.7.0 release does not support `#heuristic` directives, and the main upstream branch evaluates aggregates statically. We therefore use the authors' research Qh build, which implements dynamic aggregates over partial assignments.[1] All Alpha encodings and benchmark instances

[1] `https://github.com/tilmanni/Alpha/tree/domspec_heuristics_extended_prolog`, the build referenced by [9].

are taken unmodified from that same repository: BSP and PUP from its `Evaluation` directory, the HRP encoding from the test resources of [8] it also carries.

We evaluate Alpha under two configurations on all benchmark families: `alpha_qh` (with domain heuristics enabled via `-uqh`) and `alpha_noheur` (the baseline solver without heuristics, `-ids`). On HRP, we also run `alpha_dom`, which evaluates domain heuristics natively without the Prolog bridge. In Section 4.12, we compare these runs against our native clingo variants—`la` (lazy propagator), `ga` (ground unrolled simulation), and `gc_noheur` (heuristic-free baseline)—to isolate the cost of lazy grounding from the benefit of dynamic heuristic guidance.

Search counters cannot be compared across systems: Alpha's *guesses* and *backtracks* measure lazy CDNL search interleaved with dynamic rule instantiation, whereas clasp records decision *choices* and *conflicts* over a pre-ground program. Cross-system comparisons are therefore limited to external process metrics recorded by `runlim`: wall-clock time and peak resident set size (RSS). Furthermore, because Alpha runs on the Java Virtual Machine (JVM), its peak RSS includes the heap pre-allocated via `-Xmx`, providing an upper bound on operational memory rather than the state held by the algorithm itself.

3.4 Benchmark Infrastructure and Metrics

The experimental campaigns are orchestrated using `benchmark-tool` [16], the Potassco benchmarking suite designed to automate ASP experiment execution and distribute jobs across SLURM clusters. Each run executes a specific (system, setting, instance) configuration under the supervision of `runlim` [17], which samples resource consumption while enforcing hard wall-clock time and memory limits.

All clingo runs are executed with domain heuristics enabled (`-heuristic=Domain`) and search configured to find a single answer set (`-n 1`). For each run, we record total execution time, solving and grounding components, peak resident memory (RSS), and search statistics including choices and conflicts. We also track the size of the ground heuristic component, comparing materialized directives against lazy facts, along with propagator-specific overheads such as decision calls, query evaluation times, and trail synchronization costs.

Memory is measured as the peak resident set size (RSS) of the entire solver process, which for the Prolog backend includes the embedded SWI-Prolog runtime. Because the engine incurs a constant initialization overhead, lazy Prolog runs can consume more memory than ground variants on small instances; the advantage over the ground-and-solve variants appears as the instances grow.

Ground variants count directives materialized by gringo, whereas lazy variants count template facts that expand into candidates dynamically during search. The runtime effort deferred by lazy grounding is thus captured directly through the per-decision propagator metrics.

Chapter 4

Results

4.1 Experimental Setup and Dataset

All experiments were executed on dedicated HPC cluster nodes requesting 4 cores, with hard limits of 600 seconds and 30 GB of RAM per run enforced by `runlim`. Both backends were compiled on the test hardware and the version of SWI-Prolog used was 10.0.2.

The evaluation covers the benchmark suite defined in Section 3.3:

- BSP: $n \in \{10, 20, \ldots, 200\}$ (20 instances);
- PUP: `double-`N, with $N \in \{20, 40, \ldots, 200\}$ (10 instances);
- HRP: `house-`N, with $N \in \{2, 4, \ldots, 20\}$ persons (10 instances).

Across all benchmark families, timeouts represent the exclusive first failure mode (denoted $T@N$ in the summary tables), as every failing variant encounters the 600-second time limit before exhausting the memory ceiling. Memory exhaustion is confined to the BSP instances where $n \geq 180$, where base program grounding alone exceeds the 30 GB budget, beyond all the first failures for BSP.

On BSP, search trajectories between C++ and Prolog coincide identically across all solved instances. On PUP and HRP there are minor discrepancies in choices that arise solely from the tie-breaking among equal-rank candidates.

4.1.1 Campaign Overview and Failure Modes

Tables 4.1–4.3 summarize the outcome of the benchmark campaign, reporting the number of solved instances and the first failure point for each variant. In the BSP benchmark, solver failures scale predictably with instance size: once a variant fails at a given size, it also fails for all subsequent sizes, with timeouts representing the first failure mode in all cases (between $n = 70$ and $n = 170$). On PUP, however, difficulty does not scale monotonically with size: `lc` times out on `double-80` yet successfully solves `double-100`. This behavior stems from the benchmark design, which provides only a single instance per size with significant variations in hardness between neighboring sizes; Section 4.10 analyzes this effect in detail using choice counts rather than runtimes.

	native		Prolog	
Variant	**solved**	**first failure**	**solved**	**first failure**
`gc_noheur`	15/20	T@160	15/20	T@160
`gc`	13/20	T@140	13/20	T@140
`ga`	6/20	T@70	6/20	T@70
`ga_weak`	14/20	T@150	14/20	T@150
`la`	**16/20**	**T@170**	**15/20**	**T@160**
`lc`	15/20	T@160	14/20	T@150
`la_aux`	6/20	T@70	3/20	T@40
`la_co`	**20/20**	—	**20/20**	—

Table 4.1: BSP campaign overview ($n = 10..200$, 600 s / 30 GB per run).

	native		Prolog	
Variant	**solved**	**first failure**	**solved**	**first failure**
`gc_noheur`	4/10	T@100	4/10	T@100
`gc`	3/10	T@80	3/10	T@80
`ga`	**10/10**	—	**10/10**	—
`la`	**10/10**	—	**10/10**	—
`lc`	4/10	T@80 (non-mon.)	2/10	T@60

Table 4.2: PUP campaign overview (`double-`N, $N = 20..200$, 600 s / 30 GB per run). Best results in bold.

	native		Prolog	
Variant	**solved**	**first failure**	**solved**	**first failure**
`gc_noheur`	10/10	—	10/10	—
`gc`	10/10	—	10/10	—
`ga`	10/10	—	10/10	—
`la`	10/10	—	10/10	—
`lc`	8/10	T@18	7/10	T@16

Table 4.3: HRP campaign overview (`house-`N, $N = 2..20$ persons, 600 s / 30 GB per run).

The maximum instance size solved provides different insights depending on the problem of choice.

- **On BSP**, the solvability frontier alone does not separate the approaches. Variants sharing the same base encoding (`gc_noheur`, `la`, and `lc`) all stop between $n = 150$ and $n = 160$, constrained by the heavy grounding of the base program. What distinguishes them is search efficiency: `la` reduces solving time and conflicts to near zero (Sections 4.3 and 4.6).

- **On PUP**, the Alpha-like dynamic guidance lets both `ga` and `la` solve the entire range up to $N = 200$, whereas the other variants stall around $N = 80$ under search space explosion. With search no longer the bottleneck, PUP isolates the cost of the heuristic representation itself: `la` halves both grounding time and peak memory with respect to the static aggregate unrolling of `ga` (**12.7 GB** vs **24.6 GB** at $N = 200$).
- **On HRP**, the frontier is uninformative because nearly all variants solve the entire suite; the evaluation centers instead on decision quality and on the misguidance observed in `lc`.

Three further trends are visible in the tables. First, the static ground simulation (`ga`) fails on BSP already at $n = 70$, nine sizes earlier than the baseline, under the combinatorial explosion of static unrolling. Second, the native C++ and Prolog backends yield identical frontiers whenever propagator queries remain cheap; where they diverge, the Prolog backend always fails earlier due to evaluation overhead, particularly for `lc`, where the propagator is queried tens of thousands of times. Third, `lc` shows two distinct failure modes of clingo-style aggregate evaluation over partial assignments: on BSP the heuristic never activates, because its aggregate conditions are not satisfied at any decision point, whereas on HRP delayed evaluation produces misleading suggestions that degrade search by orders of magnitude (Sections 4.4 and 4.11).

4.2 Reduction of the Ground Heuristic Component

The primary goal of this lazy evaluation was to reduce memory spent on materializing heuristics during grounding. This reduction can be directly observed in the number of ground heuristic objects generated as a function of instance size n. In standard ASP encodings, directives such as:

```
1 #heuristic b(X) : x(X), not c(X), S = #sum{Y : c(Y)}, W = (n+1)*S+X. [W, true]
```

incorporate the dynamic aggregate value directly into the heuristic weight. Because the atoms `c(Y)` are choice atoms whose truth values are determined only during search, the grounder cannot compute the sum ahead of time. It must therefore instantiate a separate directive for every possible (X, S) pair. For $X \in \{1, \ldots, n\}$ and S ranging over all reachable subset sums (i.e., integers from 0 to $n(n+1)/2$), the total number of ground directives generated by the two symmetric rules for `b` and `c` is:

$$2 \cdot n \cdot \left(\tfrac{n(n+1)}{2} + 1\right) \;=\; \Theta(n^3).$$

The lazy variants never perform this expansion. The aggregate is evaluated on the current partial assignment during search, so the materialized heuristic representation stays constant at two static template facts, regardless of n.

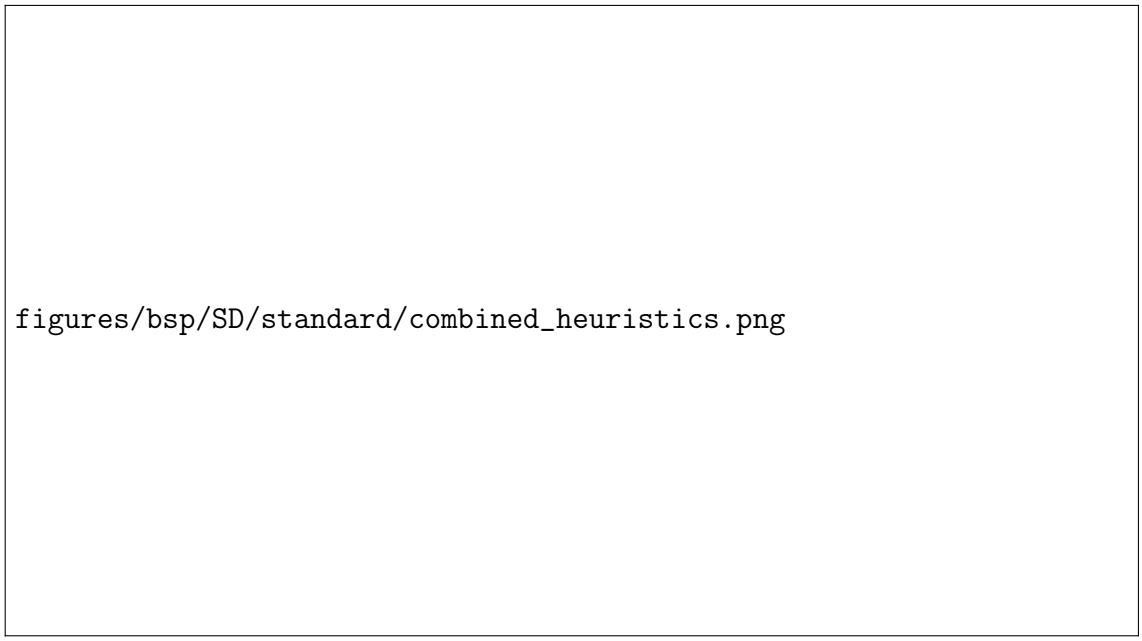

Figure 4.1: Comparison between native ground heuristic objects and lazy heuristic facts.

n	**Native heuristic objects**	**Lazy facts**	**Ratio**
10	1,120	2	560×
50	127,600	2	63,800×
100	1,010,200	2	505,100×
110	1,343,320	2	671,660×

Table 4.4: Growth of the G&S heuristic component against the lazy evaluated one.

This reduction reflects a smaller grounding footprint rather than eliminated computation: lazy evaluation shifts candidate generation to the solver's decision loop, whose runtime cost is analyzed in Section 4.6.

4.3 BSP: Runtime Behaviour

For all variants sharing the base encoding without all the grounded heuristic directives (`gc_noheur`, `la`, and `lc`), the base program grounding alone dominates overall runtime. At $n = 120$, it generates 52.8 million rules, requiring 160.9 s of grounding for `gc_noheur`, 159.7 s for `lc`, and 154.2 s for `la`.

However, a clear difference emerges here from the two readings of negation: while `gc_noheur` and `lc` require 44.1 s and 47.5 s respectively, `la` completes search in just 0.04 s. The total runtime curve for `la` therefore stays below every other variant sharing the same base encoding across the whole benchmark, reflecting the solving speedup of Figure 4.3.

Figure 4.2: One run per point; a curve ends where its variant stops solving within the limits.

Figure 4.3: `la` solving time is close to zero across all sizes, whereas `ga` degrades sharply at $n = 60$ and times out from $n = 70$ due to clasp's 16-bit weight saturation ($> 32{,}767$).

Variant	Total (s)	Grounding (s)	Solving (s)	Choices	Conflicts
`gc_noheur`	205.1	160.9	44.1	78,565	49,993
`gc`	458.2	167.7	290.5	231,260	160,115
`ga`	T	172.6	> 427.3	694,660	510,480
`ga_weak`	242.1	172.6	69.6	121,159	80,847
`la`	**154.2**	**154.2**	**0.04**	**120**	**0**
`lc`	207.2	159.7	47.5	78,565	49,993
`la_co`	0.009	< 0.01	0.01	117	0

Table 4.5: Execution time and solver statistics on BSP at $n = 120$. For `ga`, solving time represents the last value recorded before timing out.

Three points follow:

1. In `la`, default negation and dynamic aggregate evaluation let clingo use the heuristic from the first decision, taking exactly 120 choices with zero conflicts at $n = 120$.

2. `ga` matches `la` exactly up to $n = 50$, then degrades at $n = 60$ and times out from $n = 70$ onward. The cause is clasp's 16-bit weight field, as explained in Section 2.1.4: once weights saturate beyond $n = 50$, the ranking no longer discriminates between candidates.

3. `ga_weak` tracks close to the baseline and is discussed in Section 4.7, while `la_co` is analyzed in Section 4.9.

4.4 BSP: the Two Semantics Under the Microscope

As shown in Figure 4.4, `la` requires exactly n decisions with zero conflicts at every benchmarked size n. Under Alpha-like semantics, dynamic aggregate evaluation gives the propagator usable guidance from the first decision on an empty assignment, and every choice it proposes lies on the solution path.

In contrast, `lc` exhibits search statistics that are identical point-by-point to the heuristic-free baseline (`gc_noheur`), recording 78,565 choices and 49,993 conflicts at $n = 120$.

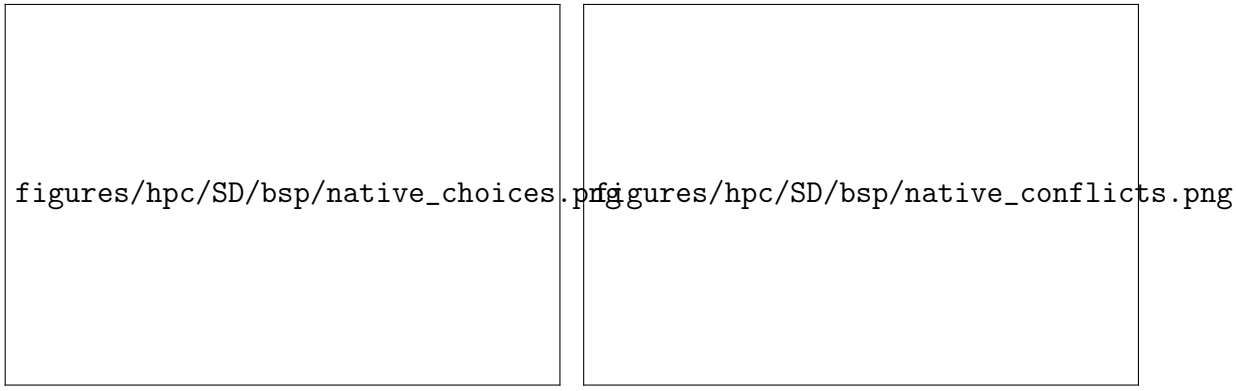

Figure 4.4: BSP search statistics on the C++ backend: solver choices (left) and conflicts (right) across sizes. `la` stays flat at n choices with zero conflicts, while `lc` overlaps the heuristic-free baseline `gc_noheur` point by point.

Under clingo semantics, aggregate guards on `c/1` require all elements to be assigned before the sum can be evaluated and an atom is considered "assigned false" only when clasp demonstrates that it cannot be true. During search, these conditions are never met at a decision point. At $n = 120$ the propagator is consulted on every decision and yields zero applicable candidates across the entire run. Because both variants run on the same native C++ backend, the comparison shows that lazy evaluation alone is not sufficient: if the semantics cannot read partial information, the solver falls back on default CDNL branching.

4.5 BSP: Rules, Variables, and Memory

The reduction of the heuristic footprint is also evident in the size of the propositional formula generated for the solver. As shown in Table 4.6, at $n = 100$ the two lazy variants replace the 1,010,200 ground directives of `gc` with the two static facts of the DSL, and their variable count (20,300) coincides exactly with the heuristic-free baseline: the heuristic leaves no trace in the propositional formula. The ground variants pay this cost in different forms. In `gc` the directives dominate the formula itself, which grows to more than one million variables, fifty times the baseline; in `ga`

the variable count stays close to the baseline, but the same million directives still have to be grounded, and beyond $n = 50$ this effort does not translate into usable guidance. Only `la_co` falls below the baseline, with 101 variables, because it also reformulates the balance constraint (Section 4.9).

Variant	Ground Directives / Facts	Solver Variables
`gc`	1,010,200	1,030,606
`ga`	1,010,200	20,402
`ga_weak`	1,010,200	20,406
`lc`	2	20,300
`la`	2	20,300
`gc_noheur`	0	20,300
`la_co`	2	101

Table 4.6: Heuristic grounding footprint and propositional solver variables at $n = 100$, C++ backend.

Figure 4.5: BSP: peak resident set size on the C++ backend across instance sizes.

Figure 4.5 shows that on large instances, memory usage is dominated by the base problem encoding; thus, while lazy heuristics eliminate the memory overhead of heuristic directives, scaling on BSP remains bounded by the ground footprint of the base program, a limitation addressed by constraint reformulation in Section 4.9.

4.6 The Price of Laziness: Per-Decision Cost

The lazy evaluation model represents a space-time trade-off where instead of suffering a combinatorial cost in memory during the grounding phase, there is a cost in computation incurred for each decision made during the search phase. There is also a remarkable difference between the two backends; the C++ one manages to be orders of magnitude faster in its decisions.

		Decide calls		**Cost (ms/decide)**		**Prolog backend only**		
Family	**Var.**	**C++**	**Prolog**	**C++**	**Prolog**	**cand./call**	**sync (s)**	**prop. share**
BSP n=120	`la`	120	120	$1.0{\cdot}10^{-4}$	0.62	121	3.4 ms	0.05%
BSP n=120	`lc`	78,565	78,565	$1.5{\cdot}10^{-4}$	0.21	0.0	3.30 s	8.7%
PUP N=160	`la`	217	197	$3.8{\cdot}10^{-4}$	3.84	5.3	8.20 s	7.1%
PUP N=200	`la`	272	249	$5.6{\cdot}10^{-4}$	6.74	5.2	16.53 s	7.3%
HRP N=20	`la`	122	106	$1.3{\cdot}10^{-4}$	9.68	1,248	0.035 s	68.3%
HRP N=14	`lc`	66,717	73,143	$0.9{\cdot}10^{-4}$	3.44	239	11.09 s	99.4%

Table 4.7: Per-decision overhead on representative solved instances. The last three columns have no C++ counterpart: the Prolog backend re-evaluates the heuristic query from scratch at every decision and mirrors the trail explicitly, whereas the C++ backend keeps its ranked targets incrementally and inspects one entry per call by construction.

The overall overhead varies dramatically across the benchmark configurations:

- When the heuristic has **perfect guidance (`la` on BSP)**, with only 120 decide calls taking 0.62 ms each in Prolog (and 10^{-4} ms in C++), propagator overhead accounts for less than 0.10% of the total runtime.
- When the semantics gets in the way **(`lc` on BSP)**, although individual query costs remain low, making many futile calls causes the propagator to consume a significant portion of the total runtime without aiding search.
- When candidate generation becomes heavy **(`la` on HRP)**, producing over 1,200 candidates per decision, Prolog evaluation time increases to 9.68 ms per call; as a result, the propagator absorbs 68% of the total runtime, even though search needs only about 100 decisions.
- When frequent search and expensive evaluation collide **(`lc` on HRP)**, making 73,143 futile calls with 239 candidates each creates a worst-case scenario: Prolog evaluation and trail synchronization eat up to 99.4% of the total runtime, two orders of magnitude above what the same misguided trajectory costs on the C++ backend.

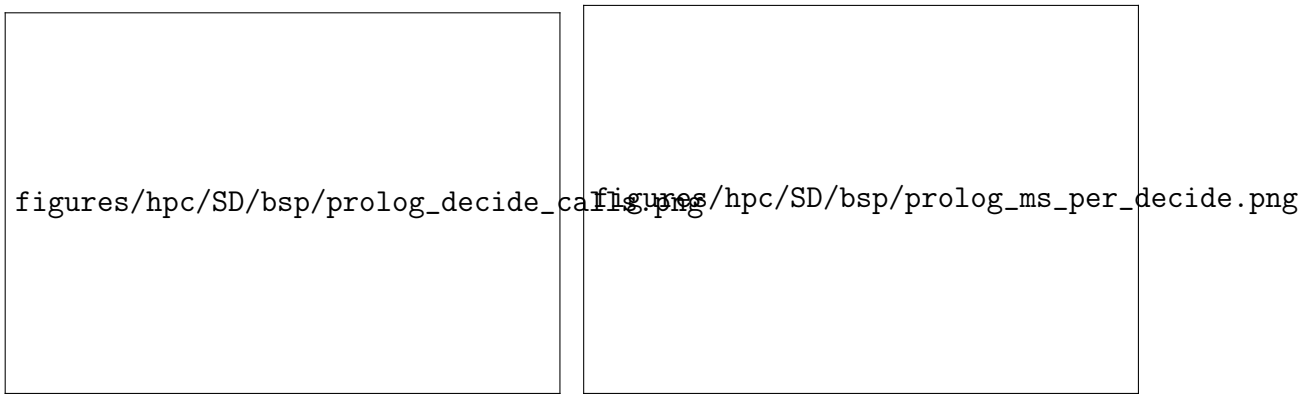

Figure 4.6: BSP, Prolog backend: decide calls and per-decision query cost. The `lc` curve pairs tens of thousands of calls with a low unit cost; `la` the opposite.

Under Alpha semantics, the query successfully identifies and evaluates all eligible candidate atoms at each step, resulting in a slightly higher query time (0.62 ms per call) due to Prolog term unification and weight calculation. Conversely, under clingo semantics, the negative body condition fails immediately on partial assignments, producing zero candidates; while this allows individual queries to terminate faster (0.21 ms), it provides no guidance whatsoever to the solver, causing a massive explosion in decision calls.

Figure 4.7: BSP: solving time comparison between the two backends. The bottom subplot shows the relative runtime ratio ($\text{Time}_{\text{Prolog}}/\text{Time}_{\text{C++}}$), where values above 1.0 indicate the factor by which the interpreted Prolog backend is slower than compiled C++.

The total lazy overhead therefore factorizes as:

Total Overhead = Decide Calls × (Query Cost + Amortized Synchronization Cost).

Lazy evaluation is thus cheapest exactly when the heuristic works: a richer condition costs more per decision, but a heuristic that guides well leaves few decisions to pay for.

4.7 Two Ground Simulations of Alpha: `ga` versus `ga_weak`

How closely can Alpha semantics be simulated within a ground-and-solve pipeline? Section 2.1.1 introduced two distinct transformations: `ga_weak`, which simply removes negative atoms from rule bodies, and `ga`, which replaces aggregate equality checks with dynamically unrolled monotone bounds ($S \leq \#\texttt{sum}\{Y : c(Y)\}$).

In BSP, both `ga` and `ga_weak` produce the exact same $\Theta(n^3)$ ground directives as the baseline `gc` (1,120 directives at $n = 10$, 8,440 at $n = 20$, and 27,960 at $n = 30$). Dropping the `sum_value(S)` domain in `ga_weak` does not reduce the grounding size: with the equality binding in place, the grounder must still enumerate all subset sums. The main difference is that equality bindings trigger at most once, and only after all atoms in the sum are assigned, whereas monotone bounds activate incrementally as the partial sum grows.

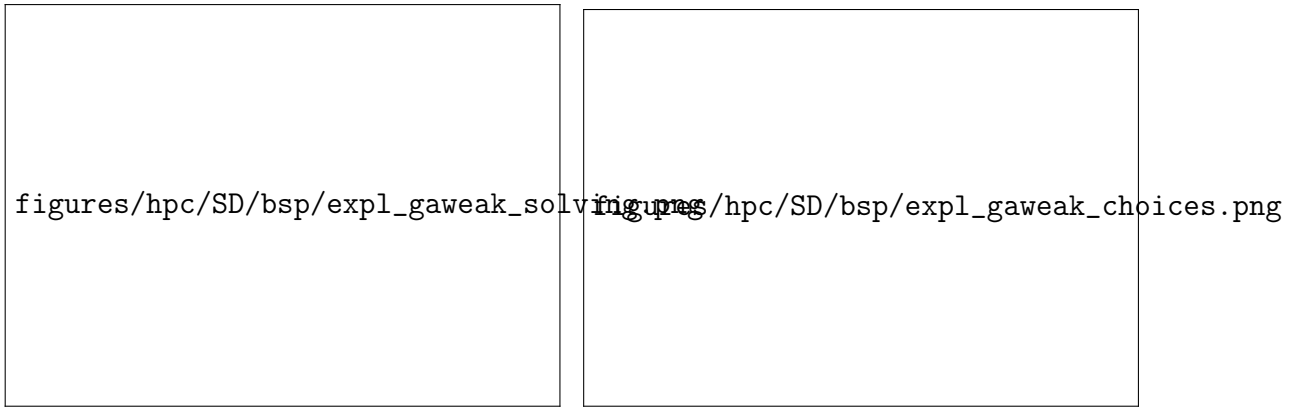

Figure 4.8: Solving time (left) and solver choices (right) for BSP comparing the Alpha-like semantics simulations (`ga` and `ga_weak`) against the baseline variants.

Beyond $n = 50$, `ga` deteriorates rapidly as weights exceed the `int16` range and saturate at 32,767: distinct partial sums collapse onto the same weight, and the ranking stops discriminating between candidate decisions (Figure 4.8). At $n = 60$ it already required 1.7 million choices and 137 s of solving time, before timing out from $n = 70$ on.

`ga_weak` never hits the timeout, even though its weight expression is the same as `ga`'s: the equality binding is satisfied only once every element of the sum is assigned, so the heuristic almost never activates and the saturation is never reached. For the same reason it gives no useful guidance. At $n = 120$ its search trajectory is worse and noisier than the heuristic-free baseline, taking about 50% more choices and correspondingly longer to solve (Table 4.5).

Omitting the negative guards without unrolling the aggregate bounds therefore yields no usable guidance: it increases search effort and slightly reduces the solvable horizon.

4.8 Evaluating Pre-Materialized Auxiliary Predicates

The `la_aux` variant investigates whether the primary advantage of lazy evaluation stems simply from reducing the number of `__heuristic` directives, or from keeping the evaluation of heuristic conditions entirely outside the ground program. In `la_aux`, heuristic conditions and aggregate values are pre-materialized into auxiliary

ASP predicates (`h_b(X,S)` and `h_c(X,S)`), which the lazy propagator then queries during search.

Defining auxiliary rules forces the grounder to instantiate the full Cartesian product of elements and partial sums (X, S). As a consequence, solver variables increase to 132,756 at $n = 50$, compared to just 5,150 in direct `la`.

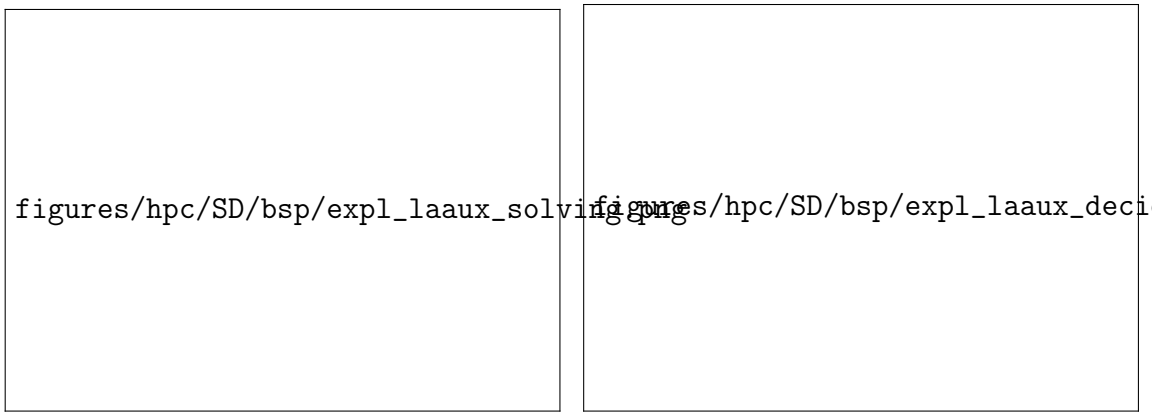

Figure 4.9: Solving time on the C++ backend (left) and number of propagator decide calls on the Prolog backend (right) comparing `la_aux` against direct `la` in BSP.

The auxiliary materialization costs performance on both backends (Figure 4.9):

- Because the propagator must scan all materialized auxiliary candidates, the per-decision cost scales as $\mathcal{O}(n^2)$. Solving time diverges sharply from $n = 40$, reaching 287.1 s at $n = 60$ before consistently timing out from $n = 70$ onward. In contrast, the direct `la` formulation maintains negligible solving times across the entire range.
- In the Prolog backend, the sheer volume of auxiliary candidates multiplies the required consultations, jumping from 573 calls at $n = 20$ to 8,452 at $n = 30$ (compared to exactly n calls in direct `la`). Trail synchronization alone accounts for 99.2% of the total runtime (383 s out of 386 s) at $n = 30$, causing the backend to time out for all sizes $n \geq 40$.

Reducing the number of heuristic directives is therefore not enough on its own: the benefit comes from keeping the evaluation of the heuristic conditions out of the ground program. Once those conditions are materialized as auxiliary rules, the combinatorial cost simply moves from the directives to the base encoding, and the lazy propagator inherits it at every decision.

4.9 Effect of the Constraint Formulation

The `la_co` variant replaces the quadratic difference check over the two sums with a single constant-bound interval constraint, substantially simplifying the grounding structure.

In the standard formulations (`gc`, `la`, and `lc`), this quadratic constraint dominates the grounding phase: at $n = 120$ it produces about 52.8 million propositional

rules and required between two and three minutes of grounding. Direct lazy evaluation (`la`) removes the heuristic directives, but the ground program remains bound by the size of the domain specification.

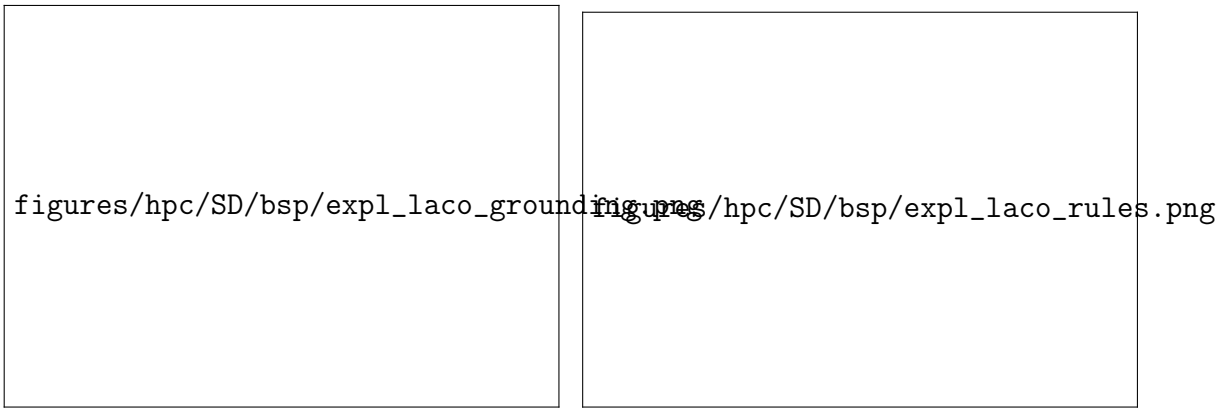

Figure 4.10: Grounding time (left) and propositional rule count (right) for BSP on the C++ backend, comparing the compact constraint formulation (`la_co`) against the baseline encodings (`gc`, `la`, and `lc`).

The reformulation changes formula size and grounding time by orders of magnitude (Figure 4.10):

- `gc`, `la`, and `lc` overlap near the top of both plots, sharing the same quadratic base encoding, whereas `la_co` flattens along the horizontal axis. At $n = 120$ the propositional formula drops from 52.8 million rules to 366.
- Grounding all but disappears. Total clingo runtime at $n = 120$ falls from 154.2 s in `la` to 9 ms, with the search guidance unchanged (117 choices, zero conflicts). The same holds up to $n = 200$, which solves in 14 ms.

The two bottlenecks are therefore independent. Once lazy heuristics remove the search cost, what remains in BSP is set entirely by the materialization of the domain constraints, and pairing lazy heuristics with a compact encoding addresses both at once.

4.10 PUP: the Grounding Bottleneck at Scale

The Partner Units Problem (PUP) is an industrial configuration benchmark on which domain-specific heuristics keep the combinatorial search under control. Unlike BSP, which uses only `#sum`, PUP combines dynamic `#count` and `#max` aggregates, the latter tracking the remaining capacity across adjacent partner units.

In BSP, unguided search dominated baseline execution times. In PUP, the dynamic Alpha-like semantics reduces solving effort to a few hundred decisions at every instance size, so more than 95% of the total execution time moves to grounding and the cost becomes one of memory. PUP therefore measures the grounding and memory efficiency of lazy evaluation at scale.

Figure 4.11 and Table 4.8 summarize the overall performance across all five variants under a 600 s timeout and a 30 GB memory limit.

Figure 4.11: Total clingo runtime on PUP across instance sizes for the C++ backend (curves terminate at the largest solved instance).

Variant	max N solved	Total (s)	Grounding (s)	Peak RSS	First failure
`gc_noheur`	80	123.2	10.6	0.7 GB	T@100
`gc`	60	97.0	17.7	1.3 GB	T@80
`ga`	200	445.1	436.6	24.6 GB	—
`la`	200	231.4	225.8	12.7 GB	—
`lc`	100	241.5	27.0	2.1 GB	T@80 (non-mon.)

Table 4.8: PUP frontier on the C++ backend (600 s / 30 GB budget): largest solved `double-`N instance and corresponding resource metrics.

Analysis of baseline failures. The baseline results reveal two distinct failure dynamics:

- In `gc_noheur`, the lack of heuristic guidance forces the solver to rely entirely on VSIDS. Combinatorial choices escalate rapidly (546,549 choices at $N = 80$), leading to a timeout at $N = 100$.
- In `gc`, the standard ground heuristic does worse than no heuristic at all, failing earlier at $N = 80$ with 865,136 choices. Because clingo's semantics requires aggregate bodies to be fully assigned before a directive is evaluated, the heuristic rules never activate during search. The inert ground directives nonetheless add variables to the solver, enlarging the search space and degrading VSIDS branching.

Grounding and memory savings (`la` vs `ga`). Both Alpha-like semantics variants (`ga` and `la`) solve the entire benchmark horizon up to $N = 200$, maintaining near-perfect search guidance (235 choices for `ga` vs 162 for `la` at $N = 120$). The two are close but not identical, and the residual difference cannot be attributed to a single cause: `ga` departs from `la` in two respects at once, having both dropped the negative guards and reconstructed the `#max` capacity terms through unrolled bounds (Section 2.1.1). Because solving takes only a few seconds, the overall performance is differentiated almost exclusively by the grounding footprint.

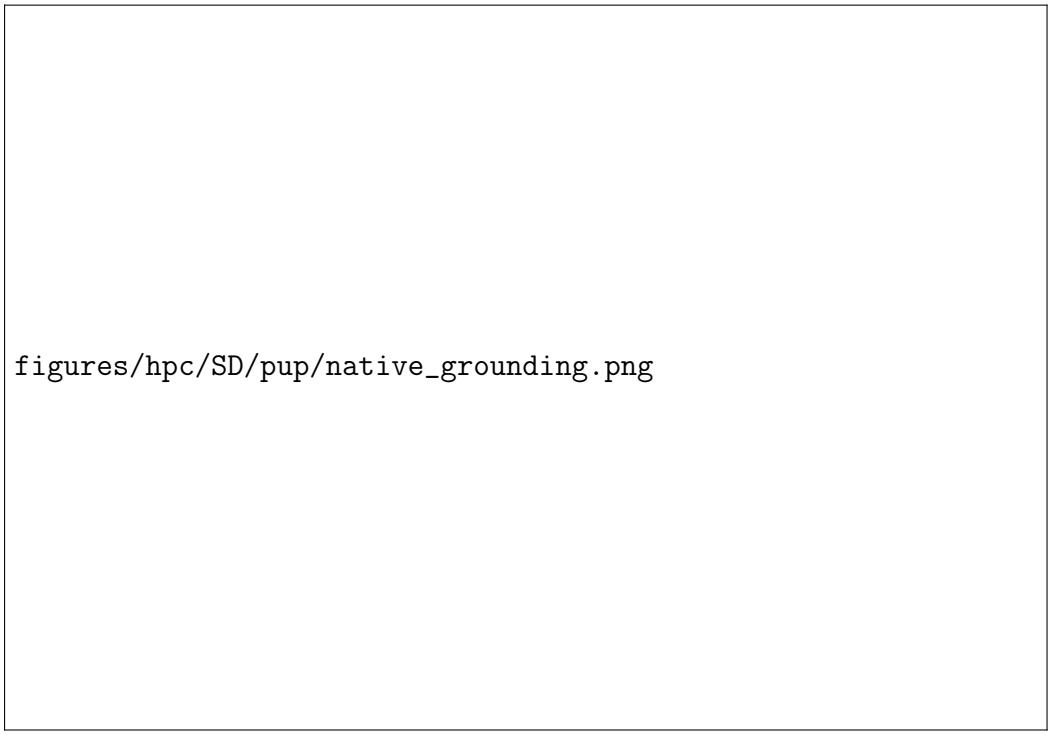

Figure 4.12: Grounding time on PUP for the C++ backend across instance sizes.

Figure 4.13: Peak memory usage on PUP for the C++ backend, illustrating the 2× memory reduction of `la` over `ga`.

As shown in Figures 4.12 and 4.13, `ga` must unroll all discrete bounds of the dynamic aggregates into the ground program. At $N = 200$ this costs 436.6 s of grounding and 24.6 GB of memory, against a 30 GB limit. Lazy evaluation in `la` leaves the aggregates to the in-memory C++ propagator and grounds only the base problem specification, reducing grounding to 225.8 s and peak memory to 12.7 GB. The saving holds across the whole range: the ratio between the two variants is largest on the small instances and settles just below a factor of two at $N = 200$, in both time and memory.

Controlled semantic comparison (`la` vs `lc`). In PUP, the lazy propagator requires auxiliary relations to expose aggregate sources. As a result, the ground propositional programs for `la` and `lc` are completely identical (216,193 rules at $N = 20$), differing exclusively in the runtime evaluation semantics configured within the propagator:

- `la` evaluates dynamic aggregates over currently true atoms, solving $N = 120$ in 47.9 s with only 162 choices.
- `lc` waits for complete assignments, generating zero active candidates during search. At $N = 120$, it makes 53,036 choices and times out.

The unguided nature of `lc` also explains its non-monotonic failure pattern in Table 4.8 (timing out on `double-80` but solving `double-100`). Without heuristic guidance, the solver is subject to the high variance of CDNL restarts, occasionally finding an early solution by chance before failing at larger sizes.

For `la`, both backends deliver guidance of the same quality, holding the search down to a few hundred choices on either engine.

Figure 4.14: Solving time (top) and Prolog-to-C++ runtime ratio (bottom) on PUP across execution backends.

As shown in Figure 4.14, solving time remains under 25 s across the whole horizon on both backends, with the Prolog runtime ratio remaining stable around 4× the native C++ engine for `la`.

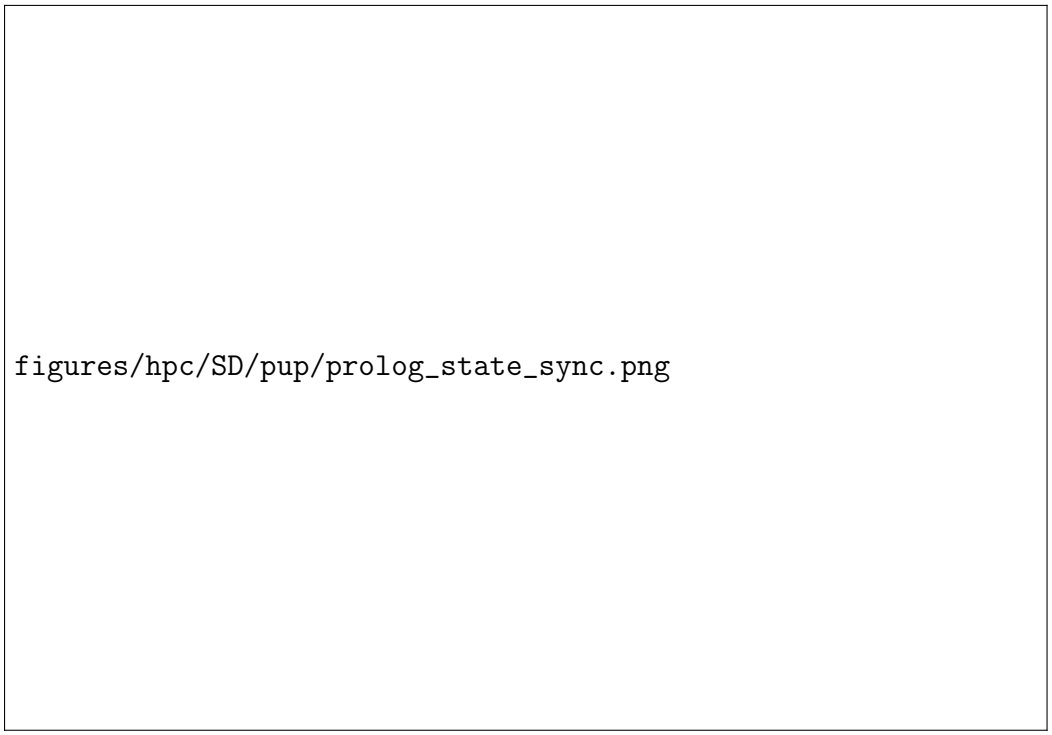

Figure 4.15: Cumulative state synchronization time for the Prolog backend on PUP.

However, as illustrated in Figure 4.15, in the Prolog backend, synchronizing the solver trail accounts for 16.5 s at $N = 200$, whereas actual Prolog query evaluation takes only 1.67 s. In comparison, the native C++ backend updates its in-process aggregate structures directly along the solver trail, completing all synchronization in just 0.68 s.

4.11 HRP: Semantics Without Aggregates

HRP evaluates heuristic semantics on rules with nested default negations but without dynamic aggregates. Because HRP instances are relatively small, nearly all variants solve every instance in under a second on the C++ backend, so the family measures guidance quality and per-decision overhead rather than reach.

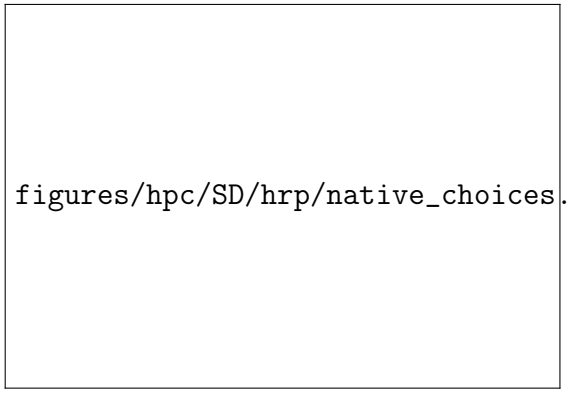

Figure 4.16: HRP: solver choices on the C++ backend (left) and the C++ versus Prolog solving comparison (right). The only diverging curve is `lc`, in both plots.

- under **Alpha-like semantics:** `la` tracks `ga` closely at every size (86 vs 88 choices at $N = 14$, with zero conflicts for both), confirming that lazy evaluation faithfully replicates reference Alpha guidance.
- under **clingo-like semantics:** the reading of negation delays the higher-level heuristic rules while letting incomplete lower-level rules fire, which misdirects the search. `lc` required 66,717 choices at $N = 14$ (an 800-fold increase over `la`) and 14.2 million choices at $N = 16$, timing out from $N = 18$.

Beyond guidance quality, HRP also illustrates the operational cost of relational candidate matching in the Prolog backend. Because the multi-room placement rules join multiple domain predicates, the SWI-Prolog engine must evaluate over 1,200 candidate pairs per decision at $N = 20$, driving unit query cost to 9.68 ms (Table 4.7).

For `la`, where search is resolved in only 106 choices, this per-decision effort accounts for 68.3% of the total runtime (1.06 s out of 1.55 s), compared to just 0.38 s on the compiled C++ backend. For `lc`, the combination of misdirected search and relational query overhead creates a catastrophic bottleneck: making 73,143 queries, already at $N = 14$, inflates total Prolog execution time to 264.5 s (with 99.4% spent inside the propagator), whereas the compiled native backend resolves the same misguided trajectory in 1.80 s.

4.12 An External Reference: Alpha

To evaluate how closely our propagator reproduces the lazy evaluation process and performance advantages, we compare our clingo variants against Alpha with and without its declarative heuristic engine (Qh mode).

On BSP, the guided Alpha configuration (`alpha_qh`) solves all twenty instances in 3.4 s and 0.37 GB at $n = 200$, requiring exactly n decisions and zero backtracks. This matches the search trajectory achieved by `la` in clingo (n choices, zero conflicts). The performance gap at larger sizes (586.5 s for `la` at $n = 160$ versus 2.4 s for `alpha_qh`,

both on the `runlim` clock) stems entirely from base program grounding in gringo (52.8 million rules at $n = 120$), which Alpha completely avoids by instantiating rules lazily along the search trail.

Figure 4.17: BSP benchmark dashboard: the four clingo variants and the heuristic-free baseline `gc_noheur` compared against Alpha with (`alpha_qh`, labeled *Alpha Qh*) and without (`alpha_noheur`, labeled *Alpha (no heur)*) declarative heuristics. Left: total wall-clock time; right: peak RSS. While `alpha_qh` stays flat along the horizontal axis, unguided `alpha_noheur` degrades rapidly, failing earlier than any clingo variant.

Lazy grounding alone is not sufficient without dynamic heuristic guidance: `alpha_noheur` solves nothing beyond $n = 40$, eleven sizes short of clingo's ground baseline `gc_noheur` ($n = 150$). The two systems sit on opposite sides of one trade-off:

- **clingo:** Because the program is fully grounded before search begins, exploring unpromising search branches is computationally very cheap: clasp only has to perform fast boolean propagation and VSIDS variable scoring over pre-existing clauses in compiled C++.
- **Alpha:** Because rules are grounded on demand during search, exploring unpromising branches is extremely expensive: every wrong decision and subsequent backtrack forces the engine to instantiate new rules and nogoods on the fly.

As shown in Figure 4.18, `alpha_noheur` suffers a severe search space explosion, peaking at over 67,000 guesses and backtracks at $n = 30$; it still solves $n = 40$, where each guess costs more even though fewer are needed, and from $n = 50$ it no longer finishes at all. With heuristic guidance, `alpha_qh` never backtracks and materializes only the minimal subset of ground rules along the single valid solution path. The two mechanisms are therefore complementary: dynamic heuristics need lazy evaluation to avoid the grounding bottleneck, and lazy grounding needs dynamic heuristics to avoid paying instantiation cost on every wrong branch.

Figure 4.18: BSP internal search counters for Alpha across instance sizes. Top row: with heuristics (`alpha_qh`), guesses remain equal to n with zero backtracks; without heuristics (`alpha_noheur`), search explodes past $n = 20$ and the curve stops at $n = 40$, the last size it solves. Bottom row: cumulative Prolog query evaluation time for `alpha_qh` (2.2 s total at $n = 200$, averaging 1.0 to 5.4 ms per consultation).

Partner Units Problem (PUP). On PUP, clingo with lazy heuristics (`la`) closely approaches Alpha's performance, solving $N = 200$ in 234.7 s and 12.7 GB compared to 50.8 s and 10.6 GB for `alpha_qh` (a factor of 4.6× in time and 1.2× in memory). In contrast, the static ground simulation `ga` required 451.4 s and 24.6 GB (nine times the runtime and over twice the memory of `alpha_qh`). Furthermore, `alpha_noheur` solves no instances on PUP, reaffirming that dynamic heuristics are essential for solving this problem family.

Figure 4.19: PUP benchmark dashboard: external comparison between clingo variants and Alpha. The baseline `alpha_noheur` does not appear as it solves zero instances. The lazy clingo variant `la` remains within a small factor of `alpha_qh` in both time and memory, whereas the ground simulation `ga` incurs substantial overhead on both axes.

Overall Frontier and Query Engines. Table 4.9 summarizes the maximal instance reach and resource consumption across systems. Regarding query engines, Alpha's Qh mode and the Prolog backend of this thesis both evaluate heuristics by querying an embedded Prolog engine over partial assignments. Consultation costs are of similar magnitude: at BSP $n = 120$, Alpha issues 242 queries at 2.87 ms each, against 120 consultations at 0.62 ms for our Prolog backend. The synchronization-based design is therefore competitive with the reference engine on consultation cost.

Family	System	max solved	Wall (s)	Peak RSS
BSP	`alpha_noheur`	40	30.5	1.1 GB
BSP	clingo `gc_noheur`	150	563.6	15.5 GB
BSP	clingo `la`	160	586.5	19.8 GB
BSP	`alpha_qh`	200	3.4	0.37 GB
BSP	clingo `la_co`	200	0.04	< 0.1 GB
PUP	`alpha_noheur`	—	—	—
PUP	clingo `gc_noheur`	80	123.3	0.7 GB
PUP	clingo `lc`	100	242.2	2.1 GB
PUP	clingo `ga`	200	451.4	24.6 GB
PUP	clingo `la`	200	234.7	12.7 GB
PUP	`alpha_qh`	200	50.8	10.6 GB

Table 4.9: Largest instance solved within the 600 s / 30 GB budget, and its cost, ordered by reach within each family. Wall time and peak RSS are `runlim` external process measures; Alpha's RSS includes the reserved JVM heap. clingo rows represent the native C++ backend. A dash marks a configuration that solved no instances.

4.13 C++ versus Prolog: Backend Comparison

Across the campaign the compiled C++ and interpreted Prolog backends behave consistently, producing identical solvability frontiers in all but five configurations. Whenever heuristic consultations are infrequent or cheap, as in `la` on PUP and HRP, both engines reach the same instance size.

Where the two backends diverge, the Prolog engine consistently fails earlier due to evaluation and synchronization overhead. On BSP, the frontier drops by a single step for `la` ($n = 160 \to 150$) and `lc` ($150 \to 140$) because both runs finish within seconds of the 600-second time limit. On PUP ($N = 100 \to 40$) and HRP ($N = 16 \to 14$) under `lc`, the unguided search triggers tens of thousands of futile queries, turning Prolog's per-call interpretation cost into the dominant bottleneck.

The C++ backend evaluates heuristic decisions three to five orders of magnitude faster than the interpreted Prolog engine ($\approx 10^{-4}$ ms against 0.2–9.7 ms per call). The native backend is therefore the faster of the two, and the Prolog backend the more expressive: its rule bodies are ordinary logic programs, which makes prototyping easier.

4.14 Summary of Results

Moving dynamic aggregate evaluation from grounding to search reduces the heuristic representation on BSP from $\Theta(n^3)$ ground directives to two constant template facts. On PUP this halves both grounding time (225.8 s against 436.6 s) and peak memory (12.7 GB against 24.6 GB) with respect to the static ground simulation `ga`. The exploratory variants show that the benefit depends on keeping the whole evaluation dynamic: omitting negative guards without unrolling the aggregates (`ga_weak`) gives no usable guidance, and pre-materializing heuristic bodies into auxiliary ASP rules (`la_aux`) puts the combinatorial explosion back into the base program. Pairing lazy heuristics with compact domain constraints (`la_co`) removes the grounding and the solving bottleneck together, solving BSP up to $n = 200$ in milliseconds.

The evaluation semantics decides whether the guidance works at all. Under Alpha-like semantics, dynamic aggregate evaluation yields usable advice on partial assignments: n choices with zero conflicts on BSP, and the intended multi-level ranking on HRP. Standard clingo semantics requires complete assignments before an aggregate body can be evaluated, which leaves the heuristic inert on BSP and misleading on HRP, where the delayed high-level rules let incomplete lower-level rules steer the solver into far larger subspaces. Static ground unrolling (`ga`) approximates dynamic aggregates on small instances, but it slows grounding and fails once the flattened weights exceed clasp's 16-bit ceiling (32,767) — a limit the lazy propagator never meets, since it ranks its decisions internally.

Lazy evaluation trades grounding memory for decision-time computation, and the total overhead is the number of decisions times the cost of a consultation. The mechanism is therefore cheapest exactly when the heuristic guides well, because the solver then makes few decisions. Against the reference lazy solver Alpha, the propagator reproduces the reference search trajectories and stays within a factor of 4.6× in execution time and 1.2× in peak memory on PUP, without the static

unrolling cost of the ground-and-solve variants.

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